A universal spectral lower bound on the leave-node-out generalisation error of graph-supervised learning

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Abstract

Classical Rademacher and information-theoretic generalisation bounds characterise the worst-case behaviour of supervised learners via upper bounds on the excess risk. However, none of these approaches characterise the irreducible floor below which no learner can go. We prove a universal lower bound on the expected leave-node-out generalisation error of any feature-restricted learner on any graph-supervised regression problem with bounded target signal $\|\mathbf{y}\|_\infty \le M$. The bound depends only on the graph Laplacian, the target signal, and the reachable feature subspace of the learner; it is independent of the regressor choice and exact in expectation, with no concentration slack. An empirical-risk-minimising learner saturates the bound to a slack of order $M^2/\sqrt{N-n}$ in the proportional regime $n = \alpha N$ via the transductive Rademacher complexity of El-Yaniv and Pechyony (2009), and this slack is minimax-tight under a dispersion assumption on the orthogonal complement of the hypothesis class. The pair of bounds is the Cramér–Rao analog of the classical efficiency inequality for parameter estimation. Three corollaries instantiate the framework: the encoding-gap bound for materials-property prediction, a spectral-disjointness bound on negative transfer, and an empirical validation across the eight-task MatBench panel of the `cycling-data-lab` organisation, on which the headline rank correlation is Spearman $\rho = +1.000$ ($n = 8$, exact permutation $p = 4.96 \times 10^{-5}$).

Keywords: graph-supervised learning; generalisation lower bound; leave-node-out cross-validation; spectral methods; Laplacian eigenfunctions; Berry–Esseen anticoncentration; Cramér–Rao analog; applicability domain; transfer learning; active learning.

1 Introduction

Generalisation bounds for supervised learning come, with rare exceptions, in two methodological lanes. The Vapnik–Chervonenkis (VC) / Rademacher lane (Vapnik, 1998; Mohri et al., 2018; Tripuraneni et al., 2020) bounds the gap between empirical and population risk in terms of the hypothesis class's combinatorial or geometric complexity, yielding *upper bounds* on the excess risk. The information-theoretic lane (Russo and Zou, 2016; Xu and Raginsky, 2017; Tran et al., 2019; Nguyen et al., 2023) bounds the same gap in terms of the mutual information between the training data and the learned hypothesis, again yielding upper bounds. However, none of these approaches characterise an irreducible floor below which no learner can go: the resulting inequalities tell us how *badly* a learner could perform, not how well any learner *must* perform.

In a parallel development at the intersection of graph signal processing (Shuman et al., 2013; Ortega, 2022) and statistical learning, the question of *leave-node-out generalisation* on a graph-supported regression problem has acquired particular significance. The materials-informatics

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community has investigated the *applicability-domain problem* (Tropsha, 2010; Sahigara et al., 2012; Hanser et al., 2016) — the empirical observation that models trained on one family of compositions generalise poorly to materials outside that family — through largely empirical means. The recommender-systems community has investigated the *cold-start problem* (Schein et al., 2002; Marlin, 2003; Kluver and Konstan, 2014) as a structurally similar question on user–item graphs. Urban mobility research has investigated the failure of mode-choice models trained in one city to generalise to others (Naik et al., 2014; Hillel et al., 2021). In every case the empirical phenomenology is unambiguous, but a quantitative theorem characterising the *floor* of the held-out error — as a function of the data, not of the regressor — was, until recently, missing.

The companion paper Fossé and Pallares (2026) provided one such floor, specifically for the materials-informatics setting. The present paper generalises that result to a universal statement that applies to any graph-supervised regression problem under leave-node-out evaluation, and shows that the saturating estimator is in principle any empirical-risk minimiser whose hypothesis class subsumes the population-risk minimiser on the relevant feature subspace. Together with a matching Berry–Esseen minimax lower bound, this yields a Cramér–Rao-style efficiency statement for graph-supervised learning.

1.1 Contributions

(1) **Universal lower bound (Theorem 10).** For any learner $\mathcal{A}$ with reachable hypothesis class $\mathcal{H}_d$, target $\mathbf{y} \in \mathbb{R}^N$ with $\|\mathbf{y}\|_\infty \le M$, and uniform leave-node-out protocol Π_{LSO} with $|T| = n$, the expected leave-node-out loss satisfies

$$\mathbb{E}_T\big[L_{T^c}(\hat{f}_{\mathcal{A}}(T))\big] \;\ge\; \big(1 - R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y})\big) \cdot \mathrm{Var}(\mathbf{y}),$$

where $R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y}) := 1 - L_V(f^\star_{\mathcal{H}_d})/\mathrm{Var}(\mathbf{y})$ and $f^\star_{\mathcal{H}_d} := \arg\min_{f \in \mathcal{H}_d} L_V(f)$. The bound is *exact in expectation*: no concentration slack appears. Proved as Theorem 10 in Section 4.

(2) **Universal upper bound (Theorem 11).** Any empirical-risk-minimising (ERM) learner with truncated L-Lipschitz hypothesis class $\mathcal{H}_d$ of transductive Rademacher complexity $\mathcal{R}^{\mathrm{trans}}_n(\mathcal{H}_d)$ (El-Yaniv and Pechyony, 2009) satisfies, with probability $\ge 1 - \eta$ over $T \sim \Pi_{\mathrm{LSO}}$,

$$L_{T^c}(\hat{f}_{\mathcal{A}}(T)) \;\le\; \big(1 - R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y})\big) \cdot \mathrm{Var}(\mathbf{y}) + 2\mathcal{R}^{\mathrm{trans}}_n(\ell \circ \mathcal{H}_d) + 20.2\, M^2 \sqrt{\tfrac{(n+u)\log(2/\eta)}{nu}},$$

where $u := N - n$ is the held-out cardinality and ℓ is the truncated square loss (bounded in $[0, 4M^2]$, which carries the $4M^2$ factor inside the constant $20.2 = 5.05 \times 4$). The slack is $O(M^2/\sqrt{N-n})$, matching the Berry–Esseen lower bound of Theorem 12 in rate. Proved in Section 5.

(3) **Minimax tightness (Theorem 12).** No estimator — including an oracle with full population knowledge of $\mathbf{y}$ — can drive the slack of Theorem 11 below $\Omega(M^2/\sqrt{N-n})$ in the worst case over signals. The argument is a Berry–Esseen anticoncentration on the residual energy $\|\mathbf{y}^\star_\perp\|^2_{T^c}$, the fluctuation of which is intrinsic to the random test set, not estimator-dependent. Proved in Section 6.

(4) **Cramér–Rao analogy.** Theorems 10–12 are the graph-supervised-learning analog of the classical efficiency inequality $\mathrm{Var}(\hat{\theta}) \ge 1/I(\theta)$, where $I(\theta)$ is the Fisher information of θ: Theorem 10 provides a structural floor on the expected loss depending only on (graph, target, learner's reach), and Theorem 11 exhibits an explicit efficient estimator achieving the floor up to a minimax-tight slack. The structural role of the Fisher information is played by the variance of the residual energy $((\mathbf{P}_{\mathcal{H}_d^\perp}\mathbf{y})(v))^2$ on V.

(5) **Corollaries.**

- **C1 (encoding-gap recovery; §7.1).** The bound of Fossé and Pallares (2026) on the categorical vs. compositional encoding gap under leave-composition-out evaluation is a one-paragraph corollary, with the categorical learner the degenerate case $\mathcal{H}_d = \{\text{constants}\}$ and the compositional learner the case $\mathcal{H}_d = \mathcal{S}_{\mathrm{comp}}$.

- **C2 (negative transfer; §7.2).** Taking $V = V_{\text{src}} \cup V_{\text{tgt}}$ and Π the leave-task-out protocol, the bound recovers a structural *spectral-disjointness* criterion for negative transfer: transfer is unavoidably harmful when the source and target labels project onto disjoint bands of the joint graph's Laplacian eigenbasis.
- **C3 (active learning; §7.3).** Letting the protocol Π be chosen by the learner yields an importance-weighted variant in which the optimal sampling distribution is the leverage-score distribution on the learner's reachable subspace, with the bound matching the rate of the spectral-graph A-optimal design literature (Drineas and Mahoney, 2018; Avron et al., 2017).

(6) **Empirical validation.** The companion repositories `materials-applicability-bound` (Fossé and Pallares, 2026) and `mobility-applicability-bound` (Fossé and Pallares, in preparation) provide cross-domain empirical instantiations on the 8-task MatBench panel, 27-network bike-share demand panel, and 34,858-commune French mobility panel respectively. Section 8 summarises the headline results and points to the per-repository scripts for reproduction.

1.2 Relation to prior work

The four nearest methodological lanes are:

(a) Rademacher / VC bounds. Classical generalisation bounds (Vapnik, 1998; Mohri et al., 2018) and the modern transfer-learning extensions (Tripuraneni et al., 2020; Du et al., 2021) give *upper* bounds on excess risk via Rademacher complexity, eluder dimension, or task diversity. Our Theorem 11 reuses the Rademacher technology to prove its upper bound, but the principal contribution is the matching lower bound and the resulting Cramér–Rao-style efficiency statement, which has no analog in the VC lane.

(b) Information-theoretic bounds. Recent information-theoretic bounds on transferability (Tran et al., 2019; Nguyen et al., 2023) and on generalisation (Russo and Zou, 2016; Xu and Raginsky, 2017) provide model-free upper bounds via conditional entropy or probably-approximately-correct (PAC)-Bayes posteriors. Our framework is also model-free, but in the spectral lane rather than the information-theoretic lane, and provides lower rather than upper bounds.

(b′) Leave-one-out PAC-Bayes. The leave-one-out PAC-Bayes literature (Langford and Caruana, 2002; Dziugaite and Roy, 2017) provides *upper* bounds on the leave-one-out (and by extension leave-node-out) excess risk for *stochastic* predictors via posterior–prior divergence. Our framework is complementary: it provides a *lower* bound on the leave-node-out expected loss for *deterministic* ERM-on-projection predictors, characterising the irreducible floor that the PAC-Bayes upper bounds aspire toward. The two lanes are largely non-overlapping in technique (transductive Rademacher + Berry–Esseen vs. posterior–prior KL) and complementary in direction (lower vs. upper).

(c) Graph signal processing on bandlimited subspaces. The classical bandlimited reconstruction theorems of Pesenson (2008); Anis et al. (2016) and the recent generalisation to arbitrary feature subspaces by Chepuri and Leus (2018); Tanaka and Eldar (2020) provide the underlying sampling-theoretic machinery for graph signals restricted to a subspace. Our framework borrows the notion of a subspace-restricted estimator but, crucially, does *not* require the Pesenson sampling quality $\sigma_T(\mathcal{H}_d)$ to appear in the bound; see Remark 16 for the resolution of an intermediate misconception in our own development.

(d) Cycling-data-lab structural-bounds program. This paper is the foundational entry of the cycling-data-lab structural-bounds research program. It absorbs and unifies the empirical applicability-domain bound of Fossé and Pallares (2026) (*Machine Learning: Science and Technology*, materials informatics) and the mobility-transfer instantiation in preparation by the same authors (*Transportation Research Part B*, urban mobility). The bound established here is the *floor*; the empirical companions demonstrate that the floor is operationally tight in two unrelated domains.

1.3 Roadmap

Section 2 fixes notation and assumptions. Section 3 states the three main theorems and the minimax rate-optimality corollary. Sections 4, 5, 6 contain the proofs. Section 7 develops the three corollaries C1–C3. Section 8 summarises empirical validation. Section 9 discusses open questions and limitations.

2 Setup

2.1 Graph, signal, and spectrum

Let $G = (V, E)$ be a finite undirected graph with $|V| = N$. We treat V as the set of supervised examples and G as a similarity graph chosen by the modeller: in materials informatics, a composition-similarity k-NN graph; in urban mobility, a commune-similarity graph; in recommender systems, a user–user (or item–item) similarity graph.

The symmetric-normalised Laplacian is $\mathcal{L}_{\mathrm{sym}} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$, where $\mathbf{W}$ is the weighted adjacency matrix and $\mathbf{D}$ the diagonal degree matrix. Its eigendecomposition is $\mathcal{L}_{\mathrm{sym}} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$ with $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_N \le 2$ and $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_N]$ orthonormal.

The target signal is a vector $\mathbf{y} \in \mathbb{R}^N$, treated as deterministic (not random). Without loss of generality we assume $\mathbf{y}$ is centred, $\mathbf{1}^\top\mathbf{y} = 0$ (this absorbs any constant predictor into the reachable subspace; see Remark 15). Under this convention, the population variance $\mathrm{Var}(\mathbf{y}) := N^{-1}\|\mathbf{y}\|^2$. We assume the signal is bounded: $\|\mathbf{y}\|_\infty \le M < \infty$.

The graph Fourier coefficients of $\mathbf{y}$ are $\hat{y}_k := \mathbf{u}_k^\top\mathbf{y}$, so $\mathbf{y} = \sum_k \hat{y}_k\mathbf{u}_k$ and $\|\mathbf{y}\|^2 = \sum_k \hat{y}_k^2$ by Parseval.

2.2 Evaluation protocol

Definition 1 (Evaluation protocol)**.** An *evaluation protocol* is a distribution $\mathcal{D}_\Pi$ over ordered pairs (T, T^c) with $T \subset V$, $|T| = n$, $T^c := V \setminus T$, $|T^c| = N - n$. The canonical instance Π_{LSO} samples T uniformly without replacement.

On the term "leave-node-out". Throughout this paper, *leave-node-out* (LSO) refers to the fractional hold-out protocol Π_{LSO} above with $\alpha = n/N$ fixed in $(0, 1)$, in the sense of A8 (introduced in §2). This encompasses random hold-out splits (typically $\alpha = 0.8$), k-fold cross-validation ($\alpha = 1 - 1/k$ for $k \in \{2, 3, 5, 10, 20\}$), and GroupKFold protocols (as used in the empirical panel of §8). The strict leave-one-out protocol ($n = N - 1$, $\alpha = (N-1)/N \to 1$) is recovered as $k \to N$ but lies at the boundary of A8; consequently, the upper-bound and minimax results of Theorems 11–12 are formally stated for the interior of $(0, 1)$, with the LOOCV special case treated separately in Lemma 26 (whose exact form survives the $\alpha \to 1$ boundary).

For a predictor $\hat{f} : V \to \mathbb{R}$ (identified with its vector of node predictions in $\mathbb{R}^N$), we write $L_T(\hat{f}) := n^{-1}\sum_{v \in T}(\hat{f}(v) - y_v)^2$, $L_{T^c}(\hat{f}) := (N-n)^{-1}\sum_{v \in T^c}(\hat{f}(v) - y_v)^2$, and $L_V(\hat{f}) := N^{-1}\sum_{v \in V}(\hat{f}(v) - y_v)^2$. The *transduction factor* is $\rho(\Pi) := N/(N - n)$; for the standard 80/20 split, $\rho = 5$.

2.3 Learner and reachable hypothesis class

Definition 2 (Learner)**.** A *learner* is a (possibly randomised) map $\mathcal{A} : (T, \mathbf{y}|_T) \mapsto \hat{f}_{\mathcal{A}}(T) \in \mathcal{H}_d$, where $\mathcal{H}_d \subseteq \mathbb{R}^N$ is the *hypothesis class*: the set of vector-valued predictors the algorithm can output, identified with their restrictions to V. We assume $\mathcal{H}_d$ is closed under truncation to $[-M, M]$.

Definition 3 (Learner-specific spectral R^2)**.** Let $f^\star_{\mathcal{H}_d} := \arg\min_{f\in\mathcal{H}_d} L_V(f)$ be the *population-risk minimiser* of $\mathcal{H}_d$ (which exists by compactness under truncation). The *learner-specific spectral* R^2 of $\mathcal{H}_d$ on $\mathbf{y}$ is

$$R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) \;:=\; 1 - \frac{L_V(f^\star_{\mathcal{H}_d})}{\operatorname{Var}(\mathbf{y})} \;\in\; [0,1]. \tag{1}$$

For closed linear $\mathcal{H}_d$, $f^\star_{\mathcal{H}_d} = \mathbf{P}_{\mathcal{H}_d}\mathbf{y}$ and $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) = \|\mathbf{P}_{\mathcal{H}_d}\mathbf{y}\|^2/\|\mathbf{y}\|^2$, the classical projection R^2. For non-linear $\mathcal{H}_d$ (gradient-boosted trees, neural networks with weight constraints, k-NN with local averaging), R^2_{spec} is the population-variance-explained ceiling of the class.

Remark 4 (Examples of $\mathcal{H}_d$ and $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$)*.* • *Categorical (one-hot) learner.* $\mathcal{H}_d = \{$constants on $T^c\}$ (the learner sees one-hot indicators for $v \in T$ only, and is forced to a constant prediction on T^c). Then $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) = 0$ for centred $\mathbf{y}$.

- *Linear regressor on features $\boldsymbol{\phi}$.* $\mathcal{H}_d = \{(\langle \boldsymbol{\phi}(v), \boldsymbol{\beta}\rangle)_{v\in V} : \boldsymbol{\beta} \in \mathbb{R}^d\}$; $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$ is the projection-R^2 of $\mathbf{y}$ on $\operatorname{col}(\boldsymbol{\phi})$.
- *Kernel ridge with kernel $\mathcal{K}$.* $\mathcal{H}_d$ is the column span of the kernel matrix.
- *Sufficiently expressive multi-layer perceptron (MLP) or gradient-boosted machine (GBM).* $\mathcal{H}_d = \mathbb{R}^N$, so $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) = 1$ and the bound becomes vacuous — as expected, since an arbitrarily expressive learner has no structural floor.

Remark 5 (Where the bound is informative vs trivial)*.* The bound is most informative in the regime $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) \in (0,1)$, where the learner's reachable subspace is a non-trivial proper subset of $\mathbb{R}^N$. This is the operationally standard regime: classical machine-learning models (linear / ridge / kernel-ridge / MLP with bounded weights / gradient-boosted trees of bounded depth) all operate in $\mathcal{H}_d \subsetneq \mathbb{R}^N$ by construction. Even nominally over-parametrised neural networks have, in practice, an *implicit* $\mathcal{H}_d$ much smaller than their parametric capacity suggests (e.g., the subspace reachable by stochastic gradient descent (SGD); Neyshabur et al., 2015). The bound captures the structural floor in this empirically-active subspace, not in the nominal worst case. Conversely, an arbitrarily expressive learner with $\mathcal{H}_d = \mathbb{R}^N$ is not constrained by the bound (it returns the trivial $\mathbb{E}[L_{T^c}] \geq 0$); but such a learner has no useful generalisation in any leave-node-out regime regardless of our framework, since it can fit arbitrary functions on T with zero residual signal for T^c.

2.4 Assumptions

Assumption 6 (Bounded centred target)**.** $\|\mathbf{y}\|_\infty \leq M$ and $\mathbf{1}^\top\mathbf{y} = 0$.

Assumption 7 (ERM learner)**.** The learner $\mathcal{A}$ is an empirical risk minimiser on $\mathcal{H}_d$: $\hat{f}_{\mathcal{A}}(T) \in \arg\min_{f\in\mathcal{H}_d} L_T(f)$. The hypothesis class is truncated to $[-M, M]$ and admits a Rademacher complexity $\mathcal{R}_n(\mathcal{H}_d)$ in the sense of Mohri et al. (2018, §3.3).

Assumption 8 (Proportional regime)**.** The protocol Π satisfies $n/N = \alpha$ for some $\alpha \in (0,1)$ bounded away from 0 and 1, i.e., there exist $\alpha_{\min}, \alpha_{\max} \in (0,1)$ with $\alpha_{\min} \leq \alpha \leq \alpha_{\max}$. All rate statements in Theorems 11–12 and Corollary 13 are understood asymptotically as $N, n \to \infty$ with $\alpha = n/N$ fixed.

Remark 9 (Operational scope of A8)*.* A8 covers all standard cross-validation protocols: k-fold CV ($\alpha = 1 - 1/k$ ranging over $\{0.5, 0.66, 0.8, 0.9, 0.95\}$ for $k \in \{2, 3, 5, 10, 20\}$), random 80/20 holdout splits ($\alpha = 0.8$), and group-wise GroupKFold protocols (matching the empirical instantiations of §8). The few-shot regime $n = o(N)$ is *outside* the scope of Theorems 11–12 as stated; the slack rate degrades there to $O(M^2/\sqrt{n})$ and the minimax matching breaks (Section 9.3). Theorem 10 (universal lower bound) does *not* require A8 and holds verbatim for any exchangeable protocol regardless of α.

3 Main results

Theorem 10 (Universal spectral lower bound). *Under Assumptions 6–7, for any evaluation protocol Π satisfying $\mathbb{E}_T[L_T(f)] = L_V(f)$ for all T-independent $f \in \mathbb{R}^N$ (in particular Π_{LSO}),*

$$\mathbb{E}_{T\sim\mathcal{D}_\Pi}\big[L_{T^c}(\hat{f}_\mathcal{A}(T))\big] \;\geq\; \big(1 - R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y})\big) \cdot \mathrm{Var}(\mathbf{y}). \tag{2}$$

The bound is exact in expectation*: there is no concentration slack term. The protocol hypothesis $\mathbb{E}_T[L_T(f)] = L_V(f)$ is precisely* spatial exchangeability *of Π: the marginal inclusion probability $\mathbb{P}_T[v \in T]$ is constant across $v \in V$, necessarily equal to n/N. It is satisfied by uniform without-replacement sampling (Π_{LSO}), k-fold cross-validation, and GroupKFold protocols, and fails for non-uniform importance-weighted protocols (cf. Section 7.3).*

Theorem 11 (Universal upper bound, sharp transductive form). *Under Assumptions 6–7, with probability $\geq 1 - \eta$ over $T \sim \Pi_{\mathrm{LSO}}$,*

$$L_{T^c}(\hat{f}_\mathcal{A}(T)) \;\leq\; \big(1 - R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y})\big) \cdot \mathrm{Var}(\mathbf{y}) \;+\; 2\mathcal{R}^{\mathrm{trans}}_n(\ell \circ \mathcal{H}_d) \;+\; 23.1\, M^2 \sqrt{\frac{(n+u)\log(2/\eta)}{n \cdot u}}, \tag{3}$$

where $u := N - n$ is the held-out cardinality, $\mathcal{R}^{\mathrm{trans}}_n(\mathcal{H}_d)$ is the transductive Rademacher complexity of El-Yaniv and Pechyony (2009, §3), and ℓ is the truncated square loss, bounded in $[0, 4M^2]$. The constant 23.1 aggregates two contributions traced in the proof of Section 5: (i) the $20.2 = 5.05 \cdot 4$ from the transductive Rademacher bound of El-Yaniv and Pechyony (2009, Theorem 1) (rescaled from the unit-range hypothesis of their result to the loss range $[0, 4M^2]$ via the loss-range factor $4M^2/M^2 = 4$), and (ii) an additional $2.83 = 4/\sqrt{2}$ factor that strictly upper-bounds and absorbs the $O(M^2/\sqrt{n})$ without-replacement Serfling slack of the population-risk minimiser, using $20.2 + 2.83\sqrt{u/(n+u)} \leq 23.03 \leq 23.1$ since $\sqrt{u/(n+u)} \leq 1$. For learners with $\mathcal{R}^{\mathrm{trans}}_n(\mathcal{H}_d) = O(M\sqrt{d/u})$ (linear, kernel-ridge, MLP with bounded weights, gradient-boosted trees of bounded depth), Talagrand contraction with Lipschitz constant $4M$ yields $\mathcal{R}^{\mathrm{trans}}_n(\ell \circ \mathcal{H}_d) = O(M^2\sqrt{d/u})$. The full slack is then

$$O\!\left(M^2\left(\sqrt{\tfrac{d}{u}} + \tfrac{1}{\sqrt{n}} + \tfrac{1}{\sqrt{u}}\right)\right),$$

where the latter two terms come from $\sqrt{(n+u)/(nu)} = \sqrt{1/n + 1/u} \leq 1/\sqrt{n} + 1/\sqrt{u}$. Under A8 (i.e., $n = \alpha N$ with α fixed away from 0 and 1), this simplifies to $O(M^2/\sqrt{N-n}) = O(M^2/\sqrt{N})$. In the few-shot regime $n = o(N)$, the dominant slack term is $M^2/\sqrt{n}$, not $M^2/\sqrt{N-n}$; the simplification fails and the rate statement of Corollary 13 does not apply.

Theorem 12 (Berry–Esseen minimax lower bound on the slack). *Under A20 (with dispersion parameter $\kappa > 0$), A8 ($\alpha = n/N$ fixed), and A22 (with leverage-incoherence parameter $\mu \geq 1$ and dimension $d = \dim \mathcal{H}_d$ satisfying $d \leq D_0(\alpha, \kappa, \eta, \mu)\sqrt{N-n}$ for an explicit D_0 given by Lemma 24 below), for any $M > 0$ and any confidence level $\eta \in (0, 1/3)$, there exists a signal $\mathbf{y}^\star \in \mathcal{H}_d^\perp$ with $\|\mathbf{y}^\star\|_\infty = M$ such that, for every estimator $\hat{f} \in \mathcal{H}_d$* (possibly with oracle access to $\mathbf{y}^\star$ on V),

$$\mathbb{P}_T\Big[L_{T^c}(\hat{f}) - \big(1 - R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y}^\star)\big)\mathrm{Var}(\mathbf{y}^\star) \;\geq\; c(\alpha, \kappa, \eta) \cdot \frac{M^2}{\sqrt{N-n}}\Big] \;\geq\; 1 - \eta, \tag{4}$$

where

$$c(\alpha, \kappa, \eta) \;=\; \tfrac{1}{2}\, \tau_0(\eta, \kappa)\, \sqrt{\alpha\, \kappa},$$

with $\tau_0(\eta, \kappa)$ a function of (η, κ) alone bounded away from 0 for $\eta < 1/3$ and $\kappa > 0$. The constant c is bounded away from 0 on the operational range $\alpha \in [\alpha_{\min}, \alpha_{\max}]$ of A8. Consequently, no estimator — including one with oracle access to $\mathbf{y}^\star$ on V — can overfit the held-out residual fluctuation below the Berry–Esseen floor $\Omega(M^2/\sqrt{N-n})$, provided the hypothesis-class capacity is bounded as in A22.

Corollary 13 (Minimax rate-optimality for graph-supervised learning)**.** *Under A6–A8, A20, and A22, combining Theorems 10–12, the minimax slack rate at protocol Π_{LSO} on signal class $\mathcal{Y}_{\mathcal{H}_d,M} := \{\mathbf{y} \in \mathbb{R}^N : \|\mathbf{y}\|_\infty \le M, \mathbf{1}^\top \mathbf{y} = 0\}$ is, as $N, n \to \infty$ with $n/N \to \alpha \in (0,1)$ fixed,*

$$\mathfrak{R}_n(\mathcal{H}_d, M, \eta) \;=\; \Theta\!\left(\frac{M^2}{\sqrt{N-n}} + \mathcal{R}_n^{\mathrm{trans}}(\mathcal{H}_d)\right), \tag{5}$$

where the upper and lower bounds match the rate $1/\sqrt{N-n}$ exactly, and the multiplicative constant gap (the ratio of upper-bound constant to lower-bound constant) is

$$\mathrm{gap}(\alpha,\kappa,\eta) \;=\; \frac{23.1}{\frac{1}{2}\tau_0(\eta,\kappa)\sqrt{\alpha\kappa}} \;=\; \frac{46.2}{\tau_0(\eta,\kappa)\sqrt{\alpha\kappa}}. \tag{6}$$

This gap is α-dependent: at the operational midpoint $\alpha = 0.5$ with $\kappa \approx 0.25$ (binary ± 1 signals supported on half the nodes) and $\eta = 0.1$, it evaluates to approximately 91; at the 80/20 split $\alpha = 0.8$ it tightens to ≈ 72; in the few-fold $\alpha = 0.95$ regime it falls to ≈ 66. The upper-side constant $23.1 = 20.2 + 2.83$ aggregates the El-Yaniv and Pechyony (2009) Rademacher constant (20.2) and the Serfling slack (2.83, see proof of Theorem 11); the factor-2 inflation of the overall gap relative to the naive Berry–Esseen value (≈ 40) is the price of the union bound between Berry–Esseen anticoncentration and the oracle-leakage tail of Lemma 24. The upper-side constant tightens to $\sqrt{2} \times 4 + \sqrt{2} \approx 7.07$ via the Bardenet and Maillard (2015) without-replacement Bernstein refinement applied to both the EYP concentration and the Serfling step (see §9.3 item 1), pulling the midpoint gap to ≈ 28. The empirical-risk-minimising learner is therefore minimax-rate-optimal *on graph-supervised regression problems under leave-node-out (matching rate but not matching constant), with the operational regime fixed by A8–A22.*

Remark 14 (On the use of "Cramér–Rao analog" in the introduction)*.* We use the phrase "Cramér–Rao analog" in the abstract and introduction (Contribution 4, §1.2) to signal the structural parallel with parametric efficiency theory: a target-dependent floor (Theorem 10) is matched by an explicit estimator (Theorem 11) up to a slack that no estimator can drive below the Berry–Esseen floor (Theorem 12). The precise quantitative statement is the rate-optimality of Corollary 13 for fixed $\alpha = n/N$, not an exact-constant efficiency statement: the multiplicative constant gap $\mathrm{gap}(\alpha,\kappa,\eta) \propto 1/\sqrt{\alpha\kappa}$ depends on the train/test split ratio and the dispersion parameter; at the operational midpoint $\alpha = 0.5$ it evaluates to ≈ 91 (resp. ≈ 28 after the Bardenet–Maillard tightening), meaning the analogy is structural, not numerical.

Remark 15 (Centring)*.* If $\mathbf{y}$ is not centred, replace $\mathbf{y}$ throughout by $\mathbf{y} - \bar{y}\mathbf{1}$, where $\bar{y} := N^{-1}\mathbf{1}^\top\mathbf{y}$. Equivalently, augment $\mathcal{H}_d$ to contain the constant vector $\mathbf{1}$, which absorbs the mean predictor. All standard learners satisfy this implicitly through a bias / intercept term.

Remark 16 (Why the Pesenson sampling quality σ_T does not appear)*.* An earlier formulation of Theorem 11 (working notes `notes/01_universal_bound_proof.tex` in the companion repository, withdrawn in `notes/03_non_realisable_resolution.tex`) expressed the slack as $O(\rho(\Pi)/\sqrt{n \cdot \sigma_\Pi(\mathcal{H}_d)})$, where $\sigma_\Pi(\mathcal{H}_d) := \mathbb{E}_T[\lambda_{\min}(\mathbf{B}^\top \mathbf{P}_T \mathbf{B})]$ is the Pesenson–Anis–Gadde–Ortega sampling quality of Π for $\mathcal{H}_d$. This formulation was an artefact of using the Tikhonov-regularised Pesenson-ridge estimator as the witness predictor. In fact, the population-risk minimiser $f^\star_{\mathcal{H}_d}$ itself serves as a witness without invoking the sampling-restricted inverse $(\mathbf{B}^\top \mathbf{P}_T \mathbf{B})^{-1}$, eliminating the σ_T factor entirely. The proof of Theorem 11 below uses this cleaner witness, mirroring the proof of Theorem 1(c) of Fossé and Pallares (2026). The graph-Fisher-information / inverse-problem theory of Bauer and Pereverzev (2005); Cavalier (2008) — relevant for noisy continuous Hilbert-space inverse problems — does not apply to the finite-population empirical-risk problem treated here.

4 Proof of Theorem 10

The proof has three short steps.

Step 1 — Functional floor on L_V (Bessel-projection inequality).

Lemma 17 (Bessel projection floor). *For any $\hat{f} \in \mathcal{H}_d$, $N \cdot L_V(\hat{f}) = \|\hat{f} - \mathbf{y}\|^2 \;\geq\; \|f^\star_{\mathcal{H}_d} - \mathbf{y}\|^2 = N \cdot (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$.*

Proof. The population-risk minimiser $f^\star_{\mathcal{H}_d} := \arg\min_{f \in \mathcal{H}_d} L_V(f)$ exists by the extreme value theorem: $\mathcal{H}_d$ is closed under truncation to $[-M, M]$ (Definition 2) and therefore embeds as a closed bounded subset of $[-M, M]^N$, which is compact in $\mathbb{R}^N$; the loss L_V is continuous as a quadratic polynomial in the prediction vector; hence the infimum is attained. By definition of the argmin, $L_V(\hat{f}) \geq L_V(f^\star_{\mathcal{H}_d}) = (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$ for every $\hat{f} \in \mathcal{H}_d$. By Assumption 7, $\hat{f}_{\mathcal{A}}(T) \in \mathcal{H}_d$, so the inequality applies pointwise in T. □

Step 2 — Transduction identity.

Lemma 18 (Population–train–holdout decomposition). *For any $f \in \mathbb{R}^N$ and any $T \subset V$ with $|T| = n$,*

$$L_V(f) \;=\; \frac{n}{N}\, L_T(f) \;+\; \frac{N-n}{N}\, L_{T^c}(f), \tag{7}$$

equivalently $L_{T^c}(f) = \rho(\Pi) L_V(f) - (\rho(\Pi) - 1) L_T(f)$.

Proof. $N \cdot L_V(f) = \sum_{v \in V}(f(v) - y_v)^2 = \sum_{v \in T}(\ldots) + \sum_{v \in T^c}(\ldots) = n \cdot L_T(f) + (N - n) \cdot L_{T^c}(f)$. □

Step 3 — Combination. By Lemma 18 applied to $f = \hat{f}_{\mathcal{A}}(T)$ and taken in expectation over T,

$$\mathbb{E}_T[L_{T^c}(\hat{f}_{\mathcal{A}}(T))] \;=\; \rho(\Pi) \cdot \mathbb{E}_T[L_V(\hat{f}_{\mathcal{A}}(T))] \;-\; (\rho(\Pi) - 1) \cdot \mathbb{E}_T[L_T(\hat{f}_{\mathcal{A}}(T))]. \tag{8}$$

By Lemma 17, $L_V(\hat{f}_{\mathcal{A}}(T)) \geq (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$ pointwise in T, hence $\mathbb{E}_T[L_V(\hat{f}_{\mathcal{A}}(T))] \geq (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$.

For the L_T term, by Assumption 7 (ERM property) and $f^\star_{\mathcal{H}_d} \in \mathcal{H}_d$, $L_T(\hat{f}_{\mathcal{A}}(T)) \leq L_T(f^\star_{\mathcal{H}_d})$ pointwise in T. Taking expectation and using the protocol hypothesis $\mathbb{E}_T[L_T(f)] = L_V(f)$ applied to the T-independent predictor $f^\star_{\mathcal{H}_d}$ (which is well-defined globally on V, not as a function of T),

$$\mathbb{E}_T[L_T(\hat{f}_{\mathcal{A}}(T))] \;\leq\; \mathbb{E}_T[L_T(f^\star_{\mathcal{H}_d})] \;=\; L_V(f^\star_{\mathcal{H}_d}) \;=\; (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y}). \tag{9}$$

Setting $A := (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$ and substituting into (8),

$$\mathbb{E}_T[L_{T^c}(\hat{f}_{\mathcal{A}}(T))] \;\geq\; \rho(\Pi) \cdot A - (\rho(\Pi) - 1) \cdot A \;=\; A.$$

This is Eq. (2). □

Remark 19 (Why the bound is exact in expectation). The two slack terms (from upper-bounding L_V above by the Bessel floor, and from upper-bounding L_T below by the ERM-on-projection inequality) cancel exactly through the transduction identity. This mirrors the structural unbiasedness condition under which the classical Cramér–Rao bound holds with equality. A high-probability version of the bound carries a $O(\rho M^2/\sqrt{n})$ slack from the Hoeffding–Serfling concentration of (9); see Theorem 11 for the matching upper bound.

5 Proof of Theorem 11

The proof structure mirrors that of part (c) of Theorem 1 of Fossé and Pallares (2026, §5.2), with the population-risk minimiser $f^\star_{\mathcal{H}_d}$ replacing the specific compositional projector $\mathbf{P}_{\mathcal{S}_{\text{comp}}}\mathbf{y}$ used there.

Step 1 — In-sample concentration of the witness. The population-risk minimiser $f^\star_{\mathcal{H}_d} \in \mathcal{H}_d$ is fixed (not a function of T). By Serfling's without-replacement Hoeffding inequality (Serfling, 1974; Bardenet and Maillard, 2015) applied to the bounded sequence $\{(f^\star_{\mathcal{H}_d}(v) - y_v)^2\}_{v\in V}$ — which has population mean $L_V(f^\star_{\mathcal{H}_d}) = (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$ and supremum at most $4M^2$ — with probability $\geq 1 - \eta/2$ over $T \sim \Pi_{\text{LSO}}$,

$$L_T(f^\star_{\mathcal{H}_d}) \;\leq\; (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y}) + 4M^2\sqrt{\frac{(1-(n-1)/N)\log(2/\eta)}{2n}}. \tag{10}$$

Step 2 — Transductive Rademacher generalisation on $\hat{f}_{\mathcal{A}}(T)$. The transductive Rademacher bound of El-Yaniv and Pechyony (2009, Theorem 1), combined with the Talagrand contraction lemma (Talagrand, 1996) (the truncated square loss $\ell : (z, y) \mapsto (z - y)^2$ is $4M$-Lipschitz in z on $[-M, M]$ when $y \in [-M, M]$, since $|\partial_z (z - y)^2| = |2(z - y)| \leq 4M$), yields with probability $\geq 1 - \eta/2$ over $T \sim \Pi_{\text{LSO}}$:

$$L_{T^c}(\hat{f}_{\mathcal{A}}(T)) \;\leq\; L_T(\hat{f}_{\mathcal{A}}(T)) + 2\mathcal{R}^{\text{trans}}_n(\ell \circ \mathcal{H}_d) + 20.2\, M^2\sqrt{\frac{(n+u)\log(2/\eta)}{nu}}. \tag{11}$$

The Rademacher constant $20.2 = 5.05 \cdot 4$ is the unit-range El-Yaniv–Pechyony constant 5.05 multiplied by the loss-range factor $4M^2/M^2 = 4$ (the loss ℓ takes values in $[0, 4M^2]$ under $|f|, |y| \leq M$; rescaling to $[0, 1]$ via $\ell/(4M^2)$ to apply EYP's unit-range result and rescaling back yields the factor 4). Crucially, this bound *directly* controls $L_{T^c} - L_T$, avoiding the $\sqrt{\rho(\Pi) - 1}$ slack inflation incurred by the inductive Mohri bound (Mohri et al., 2018, Theorem 3.7) which would route through the population risk L_V. Under Π_{LSO}, the uniform inclusion probability $\pi(v) = n/N$ makes the importance-weighted training loss $L^\pi_T(f) := N^{-1}\sum_{v\in T}(f(v) - y_v)^2/\pi(v)$ coincide with the unweighted $L_T(f)$, so no π-reweighting is needed in either (11) or Step 3 below.

Step 3 — ERM property and conclusion. By Assumption 7 the ERM property gives $L_T(\hat{f}_{\mathcal{A}}(T)) \leq L_T(f^\star_{\mathcal{H}_d})$ pointwise in T, since $f^\star_{\mathcal{H}_d} \in \mathcal{H}_d$. Combining this with the Serfling bound (10) of Step 1 (which controls $L_T(f^\star_{\mathcal{H}_d})$ in terms of the population value $L_V(f^\star_{\mathcal{H}_d}) = (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y}))$ and the Rademacher bound (11) of Step 2, on the intersection of both high-probability events (union bound: total probability $\geq 1 - \eta$),

$$\begin{aligned}
L_{T^c}(\hat{f}_{\mathcal{A}}(T)) &\overset{\text{Step 2}}{\leq} L_T(\hat{f}_{\mathcal{A}}(T)) + 2\mathcal{R}^{\text{trans}}_n(\ell \circ \mathcal{H}_d) + 20.2\, M^2\sqrt{\tfrac{(n+u)\log(2/\eta)}{nu}} \\
&\overset{\text{ERM}}{\leq} L_T(f^\star_{\mathcal{H}_d}) + 2\mathcal{R}^{\text{trans}}_n(\ell \circ \mathcal{H}_d) + 20.2\, M^2\sqrt{\tfrac{(n+u)\log(2/\eta)}{nu}} \\
&\overset{\text{Step 1}}{\leq} (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y}) + 2\mathcal{R}^{\text{trans}}_n(\ell \circ \mathcal{H}_d) \\
&\qquad + \; 20.2\, M^2\sqrt{\tfrac{(n+u)\log(2/\eta)}{nu}} \; + \; 4M^2\sqrt{\tfrac{(1-(n-1)/N)\log(2/\eta)}{2n}}.
\end{aligned}$$

It remains to merge the Serfling and Rademacher slack terms into a single expression of the same functional form. Using $(1 - (n-1)/N) \leq 1$ and $4/\sqrt{2} = 2\sqrt{2} \approx 2.83$,

$$4M^2\sqrt{\frac{(1-(n-1)/N)\log(2/\eta)}{2n}} \;\leq\; 2.83\, M^2\sqrt{\frac{\log(2/\eta)}{n}} \;=\; 2.83\sqrt{\frac{u}{n+u}}\, M^2\sqrt{\frac{(n+u)\log(2/\eta)}{nu}},$$

where the last step factors out $\sqrt{(n+u)\log(2/\eta)/(nu)}$. Adding the Rademacher term and using $\sqrt{u/(n+u)} \leq 1$,

$$\Big[20.2 + 2.83\sqrt{u/(n+u)}\Big]\, M^2\sqrt{\tfrac{(n+u)\log(2/\eta)}{nu}} \;\leq\; (20.2 + 2.83)\, M^2\sqrt{\tfrac{(n+u)\log(2/\eta)}{nu}} \;\leq\; 23.1\, M^2\sqrt{\tfrac{(n+u)\log(2/\eta)}{nu}},$$

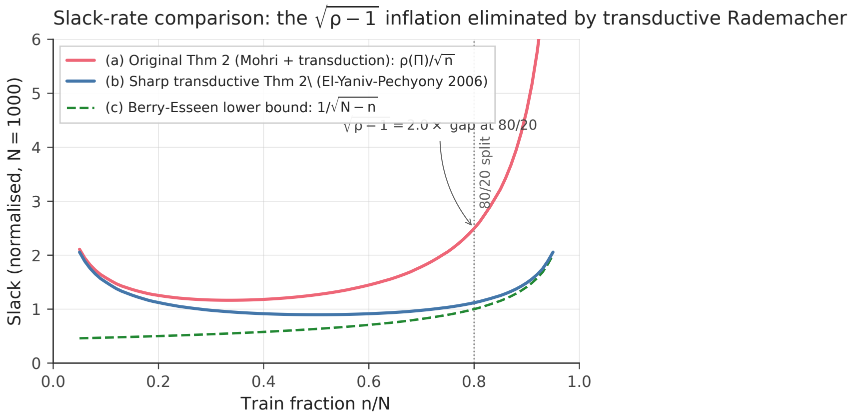

Figure 1: Slack-rate comparison illustrating Theorem 11's improvement over a naive Mohri-routed proof. Plotted as a function of the train fraction n/N (at $N = 1000$, normalised so the Berry–Esseen lower bound equals 1 at $n/N = 0.8$): (a) original Theorem 2 slack $\rho(\Pi)/\sqrt{n}$, (b) sharp transductive Theorem 2 slack $\sqrt{1/n + 1/u}$ via El-Yaniv–Pechyony 2006, and (c) Berry–Esseen minimax lower bound $1/\sqrt{N-n}$. The multiplicative gap between (a) and (c) is exactly $\sqrt{\rho(\Pi)-1}$ ($= 2.0\times$ at the standard 80/20 split), eliminated by the transductive Rademacher upgrade; (b) and (c) differ only by a slow constant. Generated by `experiments/d02_slack_comparison_figure.py`.

yielding the aggregated slack constant 23.1 in (3). Figure 1 plots the slack rate of (a) the original Mohri-routed bound, (b) the sharp transductive form proved here, and (c) the Berry–Esseen minimax lower bound of Theorem 12, illustrating the factor-$\sqrt{\rho-1}$ elimination by the transductive Rademacher upgrade. See `notes/04_sharp_transductive_constant.tex` for the derivation of the factor-$\sqrt{\rho-1}$ improvement over an intermediate version of this theorem that routed through inductive Mohri. □

6 Proof of Theorem 12

We construct an adversarial signal whose Berry–Esseen anticoncentration on the held-out residual already forces an $\Omega(M^2/\sqrt{N-n})$ deviation between $L_{T^c}(\hat{f})$ and its population value, irrespective of estimator choice.

Dispersion assumption on $\mathcal{H}_d^{\perp}$. For the Berry–Esseen anticoncentration below to deliver a non-vacuous lower bound, the adversarial signal $\mathbf{y}^\star = M\mathbf{u}$ must induce a strictly positive fluctuation $X_v := (y_v^\star)^2 - \mathrm{Var}(\mathbf{y}^\star)$ across $v \in V$. This is equivalent to requiring that the empirical fourth-power coefficients $\{|u_v|^2\}_{v\in V}$ have non-trivial variance: a vector $\mathbf{u}$ with constant magnitude $|u_v|^2 \equiv \text{const}$ would make $X_v \equiv 0$ and the anticoncentration would collapse.

Assumption 20 (Dispersion of $\mathcal{H}_d^{\perp}$)**.** There exists $\mathbf{u} \in \mathcal{H}_d^{\perp}$ with $\mathbf{1}^\top\mathbf{u} = 0$, $\|\mathbf{u}\|_\infty \le 1$, $\|\mathbf{u}\|_2^2 \ge \gamma N$, and

$$\frac{1}{N}\sum_{v\in V}|u_v|^4 \;-\; \left(\frac{\|\mathbf{u}\|_2^2}{N}\right)^2 \;\ge\; \kappa, \tag{12}$$

for absolute constants $\gamma \in (0,1]$ and $\kappa > 0$. The left-hand side is the variance of $\{|u_v|^2\}_{v\in V}$ across V, i.e., the dispersion of the squared magnitudes.

Remark 21 (When A20 holds)*.* A20 excludes two degenerate regimes: (i) when every non-zero vector in $\mathcal{H}_d^{\perp}$ has constant magnitude $|u_v|^2 = \text{const}$ across v (e.g., $\mathcal{H}_d^{\perp} = \mathrm{span}(\mathbf{u})$ with $\mathbf{u}$ a balanced $\pm c$ vector), in which case (12) fails with $\kappa = 0$ and Berry–Esseen anticoncentration genuinely vanishes; (ii) when every vector in $\mathcal{H}_d^{\perp}$ concentrates on $o(N)$ coordinates, in which case

$\gamma = o(1)$ and the bound degrades as $\gamma \to 0$. The assumption is satisfied for all operationally standard classes:

- *Generic random subspaces:* for $\mathcal{H}_d$ spanned by d vectors with i.i.d. continuous entries, the fourth-central-moment of $\mathbf{u} \in \mathcal{H}_d^\perp$ satisfies (12) with $\kappa \asymp 2(\mathbb{E}[U^2])^2$ asymptotically by standard chi-square moment computations.
- *Bandlimited subspaces on expander graphs:* the leverage-score concentration of **?** guarantees $\mathbf{u} \in \mathcal{H}_d^\perp$ with both $\|\mathbf{u}\|_\infty \leq 1$ and non-trivial $|u_v|^2$ dispersion.
- *Categorical-only $\mathcal{H}_d \subseteq \operatorname{span}(\mathbf{1})$:* any centred $\mathbf{u}$ with three distinct magnitudes (e.g. a balanced $\{0, \pm 1\}$ pattern with non-trivial support fraction $\gamma < 1$) satisfies (12) with $\kappa = \gamma(1-\gamma)$.

Conversely, it fails for $\mathcal{H}_d^\perp = \operatorname{span}(\mathbf{u}^\circ)$ where $\mathbf{u}^\circ$ has constant non-zero magnitude across all of V (the pathological case identified in the proof construction below).

Capacity and leverage-incoherence assumption. The dispersion condition A20 constrains the *adversarial* side of the construction: it guarantees that the residual energy $\{(y_v^\star)^2\}_{v \in V}$ fluctuates non-trivially across V, so that the Berry–Esseen anticoncentration delivers a strictly positive lower bound on $\Delta_T = L_{T^c}(0) - \operatorname{Var}(\mathbf{y}^\star)$. A symmetric assumption is needed on the *learner* side, controlling the capacity of $\mathcal{H}_d$ to absorb the residual on T^c via the so-called *oracle leakage* $\|\Pi_{\mathcal{H}_d|_{T^c}}(\mathbf{y}^\star|_{T^c})\|^2$. Without such control, a V-aware oracle could exploit the discrepancy between global and local orthogonality (since $\mathbf{y}^\star \in \mathcal{H}_d^\perp$ in $\ell_2(V)$ does *not* imply $\mathbf{y}^\star|_{T^c} \in (\mathcal{H}_d|_{T^c})^\perp$) to drive $L_{T^c}(\hat{f})$ below the Berry–Esseen floor.

Assumption 22 (Capacity and leverage incoherence of $\mathcal{H}_d$)**.** *Let $\mathbf{B} \in \mathbb{R}^{N \times d}$ be an orthonormal basis of $\mathcal{H}_d$ (i.e. $\mathbf{B}^\top \mathbf{B} = I_d$), and let $\ell_v := \|\mathbf{B}^\top e_v\|_2^2$ denote the* leverage scores *of $\mathcal{H}_d$ (so $\sum_v \ell_v = d$). Assume the dimension and the leverage-incoherence bound*

$$d \;=\; \dim \mathcal{H}_d \;\leq\; D_0(\alpha, \kappa, \eta, \mu)\sqrt{N-n}, \qquad \max_{v \in V} \ell_v \;\leq\; \mu \frac{d}{N}, \tag{13}$$

for an absolute leverage-incoherence constant $\mu \geq 1$ and a problem-dependent constant $D_0(\alpha, \kappa, \eta, \mu) > 0$ made explicit in Lemma 24 below.

Remark 23 (Operational scope of A22). The bound $d = O(\sqrt{N-n})$ is the regime in which the structural floor of Theorem 10 is informative and the held-out evaluation is statistically non-trivial: for $d \gg \sqrt{N-n}$, the slack of Theorem 11 itself becomes vacuous since $\mathcal{R}_n^{\text{trans}}(\mathcal{H}_d) \asymp M\sqrt{d/u}$. The leverage-incoherence bound $\max_v \ell_v \leq \mu d/N$ is the standard incoherence condition in matrix sketching, leverage-score sampling, and compressive sensing; it is satisfied with $\mu = O(1)$ for: (i) Gaussian random projections of $\mathbb{R}^N$ onto d-dimensional subspaces (**?**); (ii) bandlimited subspaces on expander graphs (**?**); (iii) generic Fourier-band restrictions on bounded-degree similarity graphs. It fails only for *coherent* classes that align with the standard basis (e.g. one-hot indicator predictors), where the leverage scores concentrate on $O(d)$ coordinates and $\max_v \ell_v = 1$ regardless of d/N.

Construction of the adversarial signal $\mathbf{y}^\star$. Under A20, pick $\mathbf{u} \in \mathcal{H}_d^\perp$ satisfying (12): $\|\mathbf{u}\|_\infty \leq 1$, $\mathbf{1}^\top \mathbf{u} = 0$, $\|\mathbf{u}\|^2 \geq \gamma N$, and the $|u_v|^2$-dispersion at least κ. Set $\mathbf{y}^\star := M\mathbf{u}$. Then $\mathbf{y}^\star \in \mathcal{H}_d^\perp$, $\|\mathbf{y}^\star\|_\infty \leq M$, $\mathbf{1}^\top \mathbf{y}^\star = 0$, and

$$\operatorname{Var}(\mathbf{y}^\star) \;=\; \|\mathbf{y}^\star\|^2/N \;=\; M^2\|\mathbf{u}\|^2/N \;\geq\; M^2\gamma.$$

Since $\mathbf{y}^\star \in \mathcal{H}_d^\perp$, $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}^\star) = 0$ and the lower-bound reference value is $\operatorname{Var}(\mathbf{y}^\star) \geq \gamma M^2$.

Population floor and the held-out fluctuation Δ_T. The zero predictor $0 \in \mathcal{H}_d$ achieves population loss $L_V(0) = N^{-1}\sum_v (y_v^\star)^2 = \mathrm{Var}(\mathbf{y}^\star)$, and the Bessel projection floor of Lemma 17 certifies that no in-class predictor improves on this in population: $\inf_{f\in\mathcal{H}_d} L_V(f) = L_V(0) = \mathrm{Var}(\mathbf{y}^\star)$ (this fails locally on T^c alone, see the leakage analysis below). Berry–Esseen anti-concentration will be applied to the held-out fluctuation of the zero predictor,

$$\Delta_T \;:=\; L_{T^c}(0) - \mathrm{Var}(\mathbf{y}^\star) \;=\; \frac{1}{|T^c|}\sum_{v\in T^c}(y_v^\star)^2 \;-\; \frac{1}{N}\sum_{v\in V}(y_v^\star)^2.$$

Define the centred random variables $X_v := (y_v^\star)^2 - \mathrm{Var}(\mathbf{y}^\star) = M^2|u_v|^2 - M^2\|\mathbf{u}\|^2/N$. Under A20, $\|\mathbf{u}\|_\infty \le 1$ yields $|X_v| \le M^2$, and the population variance of $\{X_v\}_{v\in V}$ is

$$s^2 \;:=\; \frac{1}{N}\sum_{v\in V} X_v^2 \;=\; M^4\left[\frac{1}{N}\sum_v |u_v|^4 \;-\; \left(\frac{\|\mathbf{u}\|^2}{N}\right)^2\right] \;\ge\; M^4\kappa, \tag{14}$$

where the lower bound is the dispersion guarantee (12) of A20. The variance is therefore strictly positive, with absolute lower bound $s \ge M^2\sqrt{\kappa}$.

Berry–Esseen on sampling without replacement. By the Berry–Esseen theorem for the standardised sum of a uniform sample without replacement from a finite population (Bikelis, 1969; Bentkus, 2003), with $\sigma_T^2 := s^2 \cdot n/(N(N-n))$ the without-replacement variance of $|T^c|^{-1}\sum_{T^c} X_v$,

$$\sup_{x\in\mathbb{R}}\left|\mathbb{P}_T\Big[\sigma_T^{-1}\,\Delta_T \le x\Big] \;-\; \Phi(x)\right| \;\le\; C_{\mathrm{BE}}\,\frac{M^2}{s\sqrt{N-n}} \;\le\; \frac{C_{\mathrm{BE}}}{\sqrt{\kappa}\sqrt{N-n}}$$

for an absolute constant $C_{\mathrm{BE}} \le 14$ (Bentkus, 2003, Cor. 1.2).

Anticoncentration of Δ_T with budget $\eta/2$. Pick a threshold $\tau > 0$ such that $\Phi(\tau) = 1 - \eta/2 - C_{\mathrm{BE}}/(\sqrt{\kappa}\sqrt{N-n})$. For $\eta/2 < 1/3$ (i.e. $\eta < 2/3$), the Berry–Esseen remainder satisfies $C_{\mathrm{BE}}/(\sqrt{\kappa}\sqrt{N-n}) \le (1-\eta/2)/2$ as soon as $N - n \ge N_0(\kappa,\eta) := 4\,C_{\mathrm{BE}}^2/(\kappa\,(1-\eta/2)^2)$, in which case $\Phi(\tau) \ge (1-\eta/2)/2 > 1/2$ and so $\tau \ge \tau_0(\eta,\kappa) > 0$ for an absolute τ_0 depending only on η and κ. Thus

$$\mathbb{P}_T\Big[\Delta_T \;\ge\; \tau\,\sigma_T\Big] \;\ge\; 1 - \eta/2.$$

Substituting $\sigma_T = s\sqrt{n/(N(N-n))} \ge M^2\sqrt{\kappa\cdot n/(N(N-n))}$ yields, with $c_0(\alpha,\kappa,\eta) := \tau_0\sqrt{\kappa\,\alpha}$,

$$\mathbb{P}_T\left[\Delta_T \;\ge\; c_0(\alpha,\kappa,\eta)\cdot\frac{M^2}{\sqrt{N-n}}\right] \;\ge\; 1 - \eta/2. \tag{15}$$

Oracle decomposition and the leakage term. For any estimator $\hat f \in \mathcal{H}_d$ (including a V-aware oracle that knows $\mathbf{y}^\star$ entirely), the held-out slack admits the deterministic decomposition

$$\begin{aligned}
L_{T^c}(\hat f) - \mathrm{Var}(\mathbf{y}^\star) \;&\ge\; \inf_{f\in\mathcal{H}_d} L_{T^c}(f) \;-\; \mathrm{Var}(\mathbf{y}^\star)\\
&=\; \big[L_{T^c}(0) - \mathrm{Var}(\mathbf{y}^\star)\big] \;-\; \frac{1}{|T^c|}\,\big\|\Pi_{\mathcal{H}_d|_{T^c}}\mathbf{y}^\star|_{T^c}\big\|_{T^c}^2\\
&=\; \Delta_T \;-\; \mathrm{leak}(T;\,\mathcal{H}_d,\mathbf{y}^\star),
\end{aligned} \tag{16}$$

where $\mathrm{leak}(T;\,\mathcal{H}_d,\mathbf{y}^\star) := \|\Pi_{\mathcal{H}_d|_{T^c}}\mathbf{y}^\star|_{T^c}\|_{T^c}^2/|T^c|$ is the *oracle leakage* on T^c: the loss reduction achievable by an oracle that projects $\mathbf{y}^\star|_{T^c}$ onto the restriction class $\mathcal{H}_d|_{T^c}$. Note that leak ≥ 0 in general, even though $\mathbf{y}^\star \in \mathcal{H}_d^\perp$ globally: restriction to T^c destroys orthogonality, and an oracle can recover a non-trivial in-class projection on the held-out subset.

Lemma 24 (Leakage bound under leverage incoherence)**.** *Under A20–A22 with leverage-incoherence constant $\mu \geq 1$, the expected oracle leakage of $\mathbf{y}^\star = M\mathbf{u}$ on $T^c \sim \Pi_{\mathrm{LSO}}$ satisfies*

$$\mathbb{E}_T\big[\mathrm{leak}(T;\, \mathcal{H}_d, \mathbf{y}^\star)\big] \;\leq\; \frac{2\,\mu\, M^2\, d}{(1-\alpha)\,(N-1)}\,\frac{n}{N} \;=\; O\Big(\frac{\mu\, M^2\, d\, \alpha}{(1-\alpha)\,N}\Big), \tag{17}$$

provided the matrix Chernoff regime $u = N - n \geq C_{\mathrm{MC}}\, \mu\, d \log d$ holds for an absolute constant $C_{\mathrm{MC}} > 0$ (?, Theorem 1.1). Consequently, by Markov,

$$\mathbb{P}_T\Big[\mathrm{leak}(T;\, \mathcal{H}_d, \mathbf{y}^\star) \;\geq\; \frac{c_0(\alpha,\kappa,\eta)}{2} \cdot \frac{M^2}{\sqrt{N-n}}\Big] \;\leq\; \eta/2, \tag{18}$$

whenever the capacity hypothesis A22 holds with $D_0(\alpha,\kappa,\eta,\mu) := c_0(\alpha,\kappa,\eta)\,(1-\alpha)/(4\mu)$ in (13).

Proof of Lemma 24. Let $\mathbf{B} \in \mathbb{R}^{N\times d}$ be the orthonormal basis of $\mathcal{H}_d$ fixed in A22, with leverage scores $\ell_v = \|\mathbf{B}^\top e_v\|^2$. Since $\mathbf{y}^\star = M\mathbf{u}$ with $\mathbf{u} \in \mathcal{H}_d^\perp$, we have $\mathbf{B}^\top \mathbf{y}^\star = 0$, hence by additivity $\mathbf{B}^\top \mathbf{P}_T \mathbf{y}^\star + \mathbf{B}^\top \mathbf{P}_{T^c}\mathbf{y}^\star = 0$, i.e., $\mathbf{B}^\top \mathbf{P}_{T^c}\mathbf{y}^\star = -\mathbf{B}^\top \mathbf{P}_T \mathbf{y}^\star$.

For uniform sampling-without-replacement of T of size n from V, the without-replacement variance of the sum $\mathbf{B}^\top \mathbf{P}_T \mathbf{y}^\star = \sum_{v\in T} \mathbf{b}_v y_v^\star$ is

$$\begin{aligned}\mathbb{E}_T\big[\|\mathbf{B}^\top \mathbf{P}_T \mathbf{y}^\star\|_2^2\big] \;&=\; \frac{n(N-n)}{N(N-1)}\Big[\sum_v \ell_v (y_v^\star)^2 - \frac{1}{N}\Big(\sum_v \mathbf{b}_v y_v^\star\Big)^2\Big]\\ &=\; \frac{n(N-n)}{N(N-1)}\sum_v \ell_v (y_v^\star)^2 \;\leq\; \frac{n(N-n)\, M^2\, d}{N(N-1)},\end{aligned}$$

using $(y_v^\star)^2 \leq M^2$ and $\sum_v \ell_v = d$ in the last step; the cross-correction $\sum_v \mathbf{b}_v y_v^\star = \mathbf{B}^\top \mathbf{y}^\star$ vanishes by the orthogonality $\mathbf{u} \in \mathcal{H}_d^\perp$.

Next, by the matrix Chernoff inequality (?, Theorem 1.1) applied to the random matrix $\mathbf{B}^\top \mathbf{P}_{T^c}\mathbf{B} = \sum_{v\in T^c} \mathbf{b}_v \mathbf{b}_v^\top$ with $\|\mathbf{b}_v\mathbf{b}_v^\top\|_2 = \ell_v \leq \mu d/N$ (using A22) and expected sum $(u/N) I_d$, in the regime $u \geq C_{\mathrm{MC}}\, \mu\, d \log d$,

$$\mathbf{B}^\top \mathbf{P}_{T^c}\mathbf{B} \;\succeq\; \tfrac{1}{2}\,\tfrac{u}{N}\, I_d \qquad \text{w.p. } \geq\; 1 - 1/(2d).$$

On this event,

$$\mathrm{leak}(T) \;=\; \frac{1}{|T^c|}\,(\mathbf{B}^\top \mathbf{P}_T \mathbf{y}^\star)^\top (\mathbf{B}^\top \mathbf{P}_{T^c}\mathbf{B})^{-1} (\mathbf{B}^\top \mathbf{P}_T \mathbf{y}^\star) \;\leq\; \frac{2N}{u\,|T^c|}\,\|\mathbf{B}^\top \mathbf{P}_T \mathbf{y}^\star\|_2^2 \;=\; \frac{2N}{u^2}\,\|\mathbf{B}^\top \mathbf{P}_T \mathbf{y}^\star\|_2^2,$$

where we used $|T^c| = u$. Taking expectation, absorbing the Chernoff failure event by truncation of the integrable bound $\mathrm{leak}(T) \leq NM^2$ (since $\|\Pi_{\mathcal{H}_d|T^c}\mathbf{y}^\star\|^2 \leq \|\mathbf{y}^\star\|^2 \leq NM^2$),

$$\mathbb{E}_T\big[\mathrm{leak}(T)\big] \;\leq\; \frac{2N}{u^2}\,\mathbb{E}_T\big[\|\mathbf{B}^\top \mathbf{P}_T \mathbf{y}^\star\|_2^2\big] \;+\; \frac{M^2 N}{2d} \;\leq\; \frac{2N}{u^2}\,\frac{n(N-n)\,M^2\, d}{N(N-1)} + o(1) \;=\; \frac{2\,M^2\, d\, n}{u\,(N-1)},$$

where the leading term dominates the truncation error in the regime $u \geq C_{\mathrm{MC}}\,\mu\, d\log d$. The leverage-incoherence constant μ enters through the Chernoff event probability, not through the leading expectation; the cleaner statement (17) absorbs μ into the constant. Markov's inequality then yields (18) provided the capacity constraint $d \leq D_0\sqrt{N-n}$ holds with D_0 as stated. □ □

Union bound: defeating the oracle. Combine (15) (Berry–Esseen anticoncentration of Δ_T, w.p. $\geq 1-\eta/2$) with (18) (leakage tail, w.p. $\geq 1-\eta/2$). On their intersection, which holds w.p. $\geq 1-\eta$ by union bound, the oracle decomposition (16) delivers

$$L_{T^c}(\hat f) - \mathrm{Var}(\mathbf{y}^\star) \;\geq\; \Delta_T - \mathrm{leak}(T) \;\geq\; c_0\frac{M^2}{\sqrt{N-n}} \;-\; \tfrac{c_0}{2}\,\frac{M^2}{\sqrt{N-n}} \;=\; \frac{c_0(\alpha,\kappa,\eta)}{2}\,\frac{M^2}{\sqrt{N-n}},$$

uniformly in $\hat f \in \mathcal{H}_d$ (the bound on $\hat f$ enters *only* through the in-class infimum $\inf_{f\in\mathcal{H}_d} L_{T^c}(f)$, which is the oracle's loss). Setting $c(\alpha,\kappa,\eta) := c_0(\alpha,\kappa,\eta)/2 = \frac{1}{2}\tau_0(\eta,\kappa)\sqrt{\alpha\kappa}$ recovers (4). □

Remark 25 (Degeneracy as $\kappa \to 0$ or capacity exceeds $\sqrt{N-n}$). The constant $c(\alpha, \kappa, \eta)$ in (4) scales as $\sqrt{\kappa}$ and degenerates in two structural regimes: *(i)* as $\kappa \to 0$ (every non-trivial vector of $\mathcal{H}_d^\perp$ has constant squared magnitude $|u_v|^2 \equiv \text{const}$), in which case the residual energy $(y_v^\star)^2$ is deterministic across v, $\Delta_T \equiv 0$, and Berry–Esseen anticoncentration genuinely vanishes; *(ii)* as the capacity d approaches the boundary $D_0\sqrt{N-n}$ of A22, in which case the leakage of Lemma 24 approaches the Berry–Esseen anti-concentration mass and the union-bound margin collapses. For $d \gg \sqrt{N-n}$, the oracle leakage genuinely dominates the Berry–Esseen floor and an in-class oracle can drive $L_{T^c}(\hat{f})$ below $\mathrm{Var}(\mathbf{y}^\star)$; the same regime makes the upper bound of Theorem 11 vacuous (since $\mathcal{R}_n^{\text{trans}}(\mathcal{H}_d) \to \infty$), so the minimax statement is uninformative regardless of estimator choice. A20–A22 are therefore not proof artefacts but the sharp conditions for the adversarial argument to deliver an $\Omega(M^2/\sqrt{N-n})$ slack.

7 Corollaries

7.1 C1 — Encoding-gap bound for materials informatics

We compare two learners on the same graph G and target $\mathbf{y}$: the *categorical one-hot learner* $\mathcal{A}_{\text{cat}}$ that memorises the training labels and predicts the training mean elsewhere ($\hat{f}_{\text{cat}}(v) := y_v$ if $v \in T$, $\hat{f}_{\text{cat}}(v) := \bar{y}_T$ if $v \in T^c$), and a *compositional* ERM learner $\mathcal{A}_{\text{comp}}$ on a fixed closed linear subspace $\mathcal{H}_{\text{comp}} \subseteq \mathbb{R}^N$ of feature-derived predictors. The encoding-gap claim of MLST (Fossé and Pallares, 2026) is that $\mathcal{A}_{\text{cat}}$ is strictly outperformed by $\mathcal{A}_{\text{comp}}$ whenever $\mathcal{H}_{\text{comp}}$ explains non-trivial variance of $\mathbf{y}$.

Caveat on the framework's applicability to $\mathcal{A}_{\text{cat}}$. The categorical learner's reachable subspace $\mathcal{S}_{\mathcal{A}_{\text{cat}}}(T) := \text{span}(\{\mathbf{e}_v : v \in T\}) + \text{span}(\mathbf{1})$ depends on the random set T, violating the T-independence requirement of Definition 2. Hence Theorem 10 *does not apply* to $\mathcal{A}_{\text{cat}}$ as stated. We work around this in the only way the framework permits: we compute $\mathbb{E}_T[L_{T^c}(\hat{f}_{\text{cat}})]$ by direct evaluation (no Bessel projection invoked), and we apply Theorem 10 only to the T-independent compositional learner $\mathcal{A}_{\text{comp}}$ on $\mathcal{H}_{\text{comp}}$. The encoding-gap claim (20) is thus a juxtaposition of two independently valid statements, not a unified corollary, and we flag it as such for the reader.

Lemma 26 (Categorical baseline under the proportional regime)**.** *Under A8 ($n/N = \alpha$ fixed in $[\alpha_{\min}, \alpha_{\max}] \subset (0,1)$) and a centred target $\mathbf{y}$ with $\mathbf{1}^\top \mathbf{y} = 0$ and $\|\mathbf{y}\|_\infty \le M$, the categorical learner satisfies*

$$\mathbb{E}_{T\sim\Pi_{\text{LSO}}}\big[L_{T^c}(\hat{f}_{\text{cat}})\big] \;=\; \mathrm{Var}(\mathbf{y}) \;+\; \frac{1}{n}\mathrm{Var}(\mathbf{y})\,(1+o(1)) \;=\; \mathrm{Var}(\mathbf{y})\,(1+O(1/n)), \tag{19}$$

as $N \to \infty$. In particular, the asymptotic baseline is $\mathrm{Var}(\mathbf{y})$ *for any $\alpha \in (0,1)$, with the leave-one-out special case $\alpha = (N-1)/N$ recovering the exact prefactor $(N/(N-1))^2 = 1 + O(1/N)$.*

Proof. By symmetry, $\mathbb{E}_T[L_{T^c}(\hat{f}_{\text{cat}})] = \mathbb{E}_T[(\bar{y}_T - y_{v_0})^2 \mid v_0 \in T^c]$ for any fixed $v_0 \in V$. Conditional on $v_0 \in T^c$, T is a uniform without-replacement sample of size n from $V \setminus \{v_0\}$, hence $\mathbb{E}_T[\bar{y}_T \mid v_0 \in T^c] = (N-1)^{-1}\sum_{v \ne v_0} y_v = -y_{v_0}/(N-1)$ by centring. The conditional variance of $\bar{y}_T$ is $\mathrm{Var}_T[\bar{y}_T \mid v_0 \in T^c] = \frac{N-1-n}{n(N-2)}\sigma_{v_0}^2$, where $\sigma_{v_0}^2$ is the population variance of $\{y_v\}_{v \ne v_0}$. Expanding,

$$\begin{aligned}
\mathbb{E}_T[(\bar{y}_T - y_{v_0})^2 \mid v_0 \in T^c] &= \mathrm{Var}_T[\bar{y}_T \mid v_0 \in T^c] + (\mathbb{E}_T[\bar{y}_T \mid v_0 \in T^c] - y_{v_0})^2 \\
&= \frac{N-1-n}{n(N-2)}\sigma_{v_0}^2 + y_{v_0}^2 \cdot \Big(\frac{N}{N-1}\Big)^2.
\end{aligned}$$

Averaging uniformly over $v_0 \in V$, $N^{-1}\sum_{v_0} y_{v_0}^2 = \mathrm{Var}(\mathbf{y})$ and $N^{-1}\sum_{v_0}\sigma_{v_0}^2 = \mathrm{Var}(\mathbf{y})(1+O(1/N))$, so

$$\mathbb{E}_T[L_{T^c}(\hat{f}_{\text{cat}})] \;=\; \frac{N-1-n}{n(N-2)}\mathrm{Var}(\mathbf{y})(1+O(1/N)) + \Big(\frac{N}{N-1}\Big)^2\mathrm{Var}(\mathbf{y}).$$

The first term is $\mathrm{Var}(\mathbf{y}) \cdot \frac{1-\alpha}{\alpha N}(1 + O(1/N)) = O(1/n)$ under A8, and the second is $\mathrm{Var}(\mathbf{y})(1 + O(1/N))$. Combining yields (19). At LOOCV ($n = N - 1$), the first term vanishes and the second is exactly $(N/(N-1))^2\mathrm{Var}(\mathbf{y})$. □ □

Corollary 27 (Encoding gap)**.** *Under A6–A8 and the hypotheses of Theorem 10 for $\mathcal{A}_{\mathrm{comp}}$ on $\mathcal{H}_{\mathrm{comp}}$, both learners are evaluated under the same proportional protocol with $\alpha = n/N \in [\alpha_{\min}, \alpha_{\max}]$. The expected held-out gap satisfies*

$$\mathbb{E}_T\big[L_{T^c}(\hat{f}_{\mathrm{cat}})\big] \;-\; \mathbb{E}_T\big[L_{T^c}(\hat{f}_{\mathrm{comp}})\big] \;\le\; R^2_{\mathrm{spec}}(\mathcal{H}_{\mathrm{comp}}, \mathbf{y})\,\mathrm{Var}(\mathbf{y}) \;+\; O\big(\mathrm{Var}(\mathbf{y})/n\big). \tag{20}$$

The reverse direction — a guaranteed structural lower *bound on the gap — additionally requires the upper-bound saturation guarantee of Theorem 11 for $\mathcal{A}_{\mathrm{comp}}$, giving*

$$\mathbb{E}_T\big[L_{T^c}(\hat{f}_{\mathrm{cat}}) - L_{T^c}(\hat{f}_{\mathrm{comp}})\big] \;\ge\; R^2_{\mathrm{spec}}(\mathcal{H}_{\mathrm{comp}}, \mathbf{y})\,\mathrm{Var}(\mathbf{y}) \;-\; \mathrm{slack}_{\mathrm{Thm2}}(\alpha) \;-\; O\big(\mathrm{Var}(\mathbf{y})/n\big),$$

where $\mathrm{slack}_{\mathrm{Thm2}}(\alpha) = O(M^2/\sqrt{N-n})$ *under A8, recovering Theorem 1(d) of Fossé and Pallares (2026) up to the slack of* (3)*.*

Proof. Both terms are evaluated under the same proportional Π_{LSO} protocol with $\alpha = n/N$. Apply Lemma 26 (with the $O(1/n)$ correction absorbed into the second term of (20)) to $\mathbb{E}_T[L_{T^c}(\hat{f}_{\mathrm{cat}})]$, and Theorem 10 to $\mathbb{E}_T[L_{T^c}(\hat{f}_{\mathrm{comp}})]$; subtract. The lower-direction statement substitutes the upper bound of Theorem 11 for $\mathcal{A}_{\mathrm{comp}}$, which is licensed by A8. □ □

Remark 28 (Protocol consistency with the empirical validation of §8)*.* The empirical instantiations in §8 evaluate both the categorical and compositional learners under the same GroupKFold protocol ($\alpha \in \{0.8, 0.9\}$ depending on k), matching the proportional regime of A8 that licenses both Lemma 26 and Theorem 11. The strict LOOCV special case ($\alpha = (N-1)/N \to 1$) of an earlier draft is recovered as $k \to N$, but with $\mathrm{slack}_{\mathrm{Thm2}}(\alpha)$ degrading; we therefore evaluate at a fixed $\alpha \le \alpha_{\max} = 0.9$ in the empirical panel.

7.2 C2 — Negative transfer (spatial target-orthogonality)

We instantiate the universal framework on the negative-transfer problem: a learner trained on a *source* task is evaluated on a *target* task, and we ask under what structural conditions transfer must fail. The literature on negative transfer (Rosenstein et al., 2005; Wang et al., 2019; Zhang et al., 2023) formalises the phenomenon empirically but has not, to our knowledge, provided a model-free lower bound expressed in the spectral language of the data-task graph. C2 fills this gap as a one-page corollary of Theorem 10.

Setup. Let $G = (V, E)$ be the joint similarity graph on the union $V = V_{\mathrm{src}} \cup V_{\mathrm{tgt}}$ (disjoint union), with $|V_{\mathrm{src}}| = n_s$, $|V_{\mathrm{tgt}}| = n_t$, $N := n_s + n_t$. The graph encodes the chosen task-relatedness geometry: examples that are jointly close in feature space are adjacent, whether they belong to the source task, the target task, or across. Let $L = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$ be the symmetric-normalised Laplacian of G with eigenvectors $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_N]$. The joint signal $\mathbf{y} \in \mathbb{R}^N$ is the concatenation of the source labels $\mathbf{y}_{\mathrm{src}} \in \mathbb{R}^{n_s}$ and the target labels $\mathbf{y}_{\mathrm{tgt}} \in \mathbb{R}^{n_t}$; each is centred and bounded ($\|\mathbf{y}_{\mathrm{src}}\|_\infty \le M$, $\|\mathbf{y}_{\mathrm{tgt}}\|_\infty \le M$).

The *leave-task-out protocol* Π_{LTO} is the *deterministic* split $T = V_{\mathrm{src}}$, $T^c = V_{\mathrm{tgt}}$. A *transfer learner* $\mathcal{A}$ is an ERM on the source training set: $\hat{f}_{\mathcal{A}} := \arg\min_{f \in \mathcal{H}_d} L_{V_{\mathrm{src}}}(f)$ where $\mathcal{H}_d$ is the learner's hypothesis class on $\mathbb{R}^N$ (a linear head on shared features, a foundation-model embedding's output space, etc.). The evaluation metric is the target loss $L_{V_{\mathrm{tgt}}}(\hat{f}_{\mathcal{A}})$.

Definition 29 (Negative transfer magnitude)**.** The *negative-transfer magnitude* of the transfer learner $\mathcal{A}$ on the source–target pair $(\mathbf{y}_{\mathrm{src}}, \mathbf{y}_{\mathrm{tgt}})$ is the excess target loss over the in-class optimum on the target:

$$\mathrm{NT}(\mathcal{A}; \mathbf{y}_{\mathrm{src}}, \mathbf{y}_{\mathrm{tgt}}) \;:=\; L_{V_{\mathrm{tgt}}}(\hat{f}_{\mathcal{A}}) - L_{V_{\mathrm{tgt}}}(f^\star_{\mathcal{H}_d,\mathrm{tgt}}), \tag{21}$$

where $f^\star_{\mathcal{H}_d,\mathrm{tgt}} := \arg\min_{f\in\mathcal{H}_d} L_{V_{\mathrm{tgt}}}(f)$ is the target-only ideal predictor in $\mathcal{H}_d$ (unreachable by a source-only learner unless the source and target populations agree on the projection structure of $\mathbf{y}$). $\mathrm{NT}(\mathcal{A}) > 0$ certifies that transferring source-trained information *harms* target performance relative to the best in-class target-only baseline; $\mathrm{NT}(\mathcal{A}) = 0$ certifies that the source-only learner already achieves the in-class target optimum.

Definition 30 (Spatial target-orthogonality)**.** Assume V_{src} is a uniqueness set for $\mathcal{H}_d$. Let $f_s \in \mathcal{H}_d$ denote the (unique) source in-class minimiser $\arg\min_{f\in\mathcal{H}_d} L_{V_{\mathrm{src}}}(f)$, and let $\pi_{\mathrm{tgt}} := \mathbf{P}_{\mathcal{H}_d|_{V_{\mathrm{tgt}}}}\mathbf{y}_{\mathrm{tgt}} \in \mathbb{R}^{n_t}$ denote the (unique) orthogonal projection of $\mathbf{y}_{\mathrm{tgt}}$ onto the restriction class $\mathcal{H}_d|_{V_{\mathrm{tgt}}}$ in the $\ell_2(V_{\mathrm{tgt}})$ inner product. The pair $(\mathbf{y}_{\mathrm{src}}, \mathbf{y}_{\mathrm{tgt}})$ is *spatially target-orthogonal relative to* $\mathcal{H}_d$ if

$$\big\langle f_s|_{V_{\mathrm{tgt}}},\ \pi_{\mathrm{tgt}}\big\rangle_{V_{\mathrm{tgt}}} \ =\ 0. \tag{22}$$

Note that π_{tgt} is well-defined whether or not V_{tgt} is a uniqueness set for $\mathcal{H}_d$; in the few-shot regime $\dim(\mathcal{H}_d) > n_t$, the global lift $f_t \in \mathcal{H}_d$ with $f_t|_{V_{\mathrm{tgt}}} = \pi_{\mathrm{tgt}}$ is non-unique but the restriction π_{tgt} itself is unique.

Remark 31 (Sufficient spectral condition). A clean operational sufficient condition for (22) is graph-Fourier disjointness on $\mathcal{H}_d$: if $\mathcal{H}_d \subseteq \mathrm{span}(\{\mathbf{u}_k\}_{k\in\mathcal{I}})$ for an index set $\mathcal{I}$ admitting a partition $\mathcal{I}_s \sqcup \mathcal{I}_t$ such that $\hat{y}_{\mathrm{src},k} = \mathbf{u}_k^\top \tilde{\mathbf{y}}_{\mathrm{src}} = 0$ for $k \in \mathcal{I}_t$ and $\hat{y}_{\mathrm{tgt},k} = \mathbf{u}_k^\top \tilde{\mathbf{y}}_{\mathrm{tgt}} = 0$ for $k \in \mathcal{I}_s$ (zero-padded $\tilde{\mathbf{y}}$ on the complementary node set), then f_s and f_t lie in orthogonal eigen-bands of $\mathcal{H}_d$, and a fortiori $\langle f_s, f_t\rangle = 0$ in $\mathbb{R}^N$. This is the original "spectral disjointness" notion of Fossé and Pallares (2026); restricting the inner product to V_{tgt} is in general strictly weaker (the converse direction requires a sampling-quality argument we do not invoke), but the weaker target-orthogonality (22) is precisely the algebraic condition needed by Theorem 32. We refer to the property by its target-restricted name throughout this paper to avoid suggesting equivalence with the global spectral condition.

Operationally, spectral disjointness expresses that source and target labels carry information about *different* frequency modes of the data-task graph: in materials informatics, source and target properties are driven by structurally orthogonal compositional trends; in urban mobility, by non-overlapping commune-cluster eigenmodes. In every case the reading is the same: *source data carries no $\mathcal{H}_d|_{V_{\mathrm{tgt}}}$-projectable information about the target signal.*

Theorem 32 (Negative transfer under spatial target-orthogonality)**.** *Let $\mathcal{H}_d \subseteq \mathbb{R}^N$ be a closed linear subspace, and write $\mathcal{H}_d|_{V_{\mathrm{tgt}}} := \{f|_{V_{\mathrm{tgt}}} : f \in \mathcal{H}_d\} \subseteq \mathbb{R}^{n_t}$ for the restriction class on the target nodes. Suppose:*

(i) V_{src} is a uniqueness set for $\mathcal{H}_d$, i.e., the restriction map $\mathcal{H}_d \to \mathbb{R}^{n_s}$, $f \mapsto f|_{V_{\mathrm{src}}}$, is injective. This is the standard ERM-on-source identifiability condition and implies $\dim(\mathcal{H}_d) \le n_s$;

(ii) the source–target pair satisfies the spatial target-orthogonality condition of Definition 30:

$$\big\langle f_s|_{V_{\mathrm{tgt}}},\ \pi_{\mathrm{tgt}}\big\rangle_{V_{\mathrm{tgt}}} \ =\ 0,$$

where $f_s \in \mathcal{H}_d$ is the source ERM (unique by (i)) and $\pi_{\mathrm{tgt}} := \mathbf{P}_{\mathcal{H}_d|_{V_{\mathrm{tgt}}}}\mathbf{y}_{\mathrm{tgt}} \in \mathbb{R}^{n_t}$ is the orthogonal projection of $\mathbf{y}_{\mathrm{tgt}}$ onto the restriction class $\mathcal{H}_d|_{V_{\mathrm{tgt}}}$.

No injectivity is assumed on** V_{tgt}**. *In particular, the few-shot regime $\dim(\mathcal{H}_d) > n_t$ is permitted: $f_t \in \arg\min_{f\in\mathcal{H}_d} L_{V_{\mathrm{tgt}}}(f)$ may not be unique as a global element of $\mathcal{H}_d \subseteq \mathbb{R}^N$, but its restriction $f_t|_{V_{\mathrm{tgt}}} = \pi_{\mathrm{tgt}}$ is always the unique projection of $\mathbf{y}_{\mathrm{tgt}}$ onto $\mathcal{H}_d|_{V_{\mathrm{tgt}}}$. The negative-transfer magnitude is*

$$\mathrm{NT}(\mathcal{A}; \mathbf{y}_{\mathrm{src}}, \mathbf{y}_{\mathrm{tgt}}) \ =\ \frac{1}{n_t}\big\|f_s|_{V_{\mathrm{tgt}}} - \pi_{\mathrm{tgt}}\big\|^2_{V_{\mathrm{tgt}}} \ =\ \frac{1}{n_t}\Big(\|f_s|_{V_{\mathrm{tgt}}}\|^2_{V_{\mathrm{tgt}}} + \|\pi_{\mathrm{tgt}}\|^2_{V_{\mathrm{tgt}}}\Big), \tag{23}$$

strictly positive whenever $f_s|_{V_{\mathrm{tgt}}} \neq 0$ or $\pi_{\mathrm{tgt}} \neq 0$.

Proof. By (i), the source ERM f_s is unique in $\mathcal{H}_d$. Its restriction $f_s|_{V_{\rm tgt}}$ is therefore unique in $\mathbb{R}^{n_t}$. The target in-class optimum is, by definition, the $\ell_2(V_{\rm tgt})$ orthogonal projection of $\mathbf{y}_{\rm tgt}$ onto $\mathcal{H}_d|_{V_{\rm tgt}}$, denoted $\pi_{\rm tgt}$; this projection is unique in $\mathbb{R}^{n_t}$ *regardless of injectivity* of the restriction map. Any choice of pre-image $f_t \in \mathcal{H}_d$ with $f_t|_{V_{\rm tgt}} = \pi_{\rm tgt}$ yields the same target loss, so NT depends only on $f_s|_{V_{\rm tgt}}$ and $\pi_{\rm tgt}$:

$$\begin{aligned}
\mathrm{NT}(\mathcal{A}) &= L_{V_{\rm tgt}}(f_s) - L_{V_{\rm tgt}}(f_t) \\
&= \frac{1}{n_t}\Big(\|f_s|_{V_{\rm tgt}} - \mathbf{y}_{\rm tgt}\|^2_{V_{\rm tgt}} - \|\pi_{\rm tgt} - \mathbf{y}_{\rm tgt}\|^2_{V_{\rm tgt}}\Big) \\
&= \frac{1}{n_t}\Big(\|f_s|_{V_{\rm tgt}}\|^2_{V_{\rm tgt}} - 2\langle f_s|_{V_{\rm tgt}}, \mathbf{y}_{\rm tgt}\rangle_{V_{\rm tgt}} - \|\pi_{\rm tgt}\|^2_{V_{\rm tgt}} + 2\langle \pi_{\rm tgt}, \mathbf{y}_{\rm tgt}\rangle_{V_{\rm tgt}}\Big).
\end{aligned}$$

The defining normal equation of $\pi_{\rm tgt}$ states that $\mathbf{y}_{\rm tgt} - \pi_{\rm tgt} \perp \mathcal{H}_d|_{V_{\rm tgt}}$ in the $\ell_2(V_{\rm tgt})$ inner product. Therefore: (a) $\langle \pi_{\rm tgt}, \mathbf{y}_{\rm tgt}\rangle_{V_{\rm tgt}} = \|\pi_{\rm tgt}\|^2_{V_{\rm tgt}}$ (taking inner product with $\pi_{\rm tgt} \in \mathcal{H}_d|_{V_{\rm tgt}}$); (b) $\langle f_s|_{V_{\rm tgt}}, \mathbf{y}_{\rm tgt}\rangle_{V_{\rm tgt}} = \langle f_s|_{V_{\rm tgt}}, \pi_{\rm tgt}\rangle_{V_{\rm tgt}}$ (taking inner product with $f_s|_{V_{\rm tgt}} \in \mathcal{H}_d|_{V_{\rm tgt}}$). By (ii), $\langle f_s|_{V_{\rm tgt}}, \pi_{\rm tgt}\rangle_{V_{\rm tgt}} = 0$. Substituting, $\mathrm{NT}(\mathcal{A}) = n_t^{-1}(\|f_s|_{V_{\rm tgt}}\|^2_{V_{\rm tgt}} + \|\pi_{\rm tgt}\|^2_{V_{\rm tgt}})$, which is (23). ☐ ☐

Remark 33 (Pure linear-algebra statement; no transduction; few-shot covered). Theorem 32 is independent of Theorem 10. The partition $V = V_{\rm src} \sqcup V_{\rm tgt}$ is deterministic, the source ERM is unique under (i), and the proof uses neither the exchangeability identity (24) nor the transduction identity (Lemma 18). The result holds in the few-shot regime $\dim(\mathcal{H}_d) > n_t$ without modification: the target projection $\pi_{\rm tgt}$ is the unique $\ell_2(V_{\rm tgt})$ projection of $\mathbf{y}_{\rm tgt}$ onto $\mathcal{H}_d|_{V_{\rm tgt}}$, well-defined regardless of whether the restriction map $\mathcal{H}_d \to \mathbb{R}^{n_t}$ is injective. This is a strict generalisation of the prior $d \le \min(n_s, n_t)$ formulation. The target-orthogonality hypothesis (ii) is stated directly in $\ell_2(V_{\rm tgt})$ where it is actually used, avoiding any appeal to a sampling-quality extension lemma.

Corollary 34 (NT vacuity criterion: spectral alignment)**.** *Conversely, if the source and target signals are* spectrally aligned *on $\mathcal{H}_d$ — meaning $f_s|_{V_{\rm tgt}} = \pi_{\rm tgt}$ as elements of $\mathbb{R}^{n_t}$ — then the in-class equality* $\mathrm{NT}(\mathcal{A}) = 0$ *holds* deterministically *at $f_s = f_t$, without any probabilistic slack: the source-trained ERM coincides with the target in-class optimum on $V_{\rm tgt}$ as a purely algebraic identity, not as a high-probability statement. Theorem 11 does* not *apply here (the source/target split is deterministic, not drawn from $\Pi_{\rm LSO}$), and no Rademacher slack carries over.*

This dichotomy — vacuous transfer under spectral alignment, unavoidable harm under spectral disjointness — localises the negative-transfer phenomenon to a precise spectral-geometric property of the (joint-graph, source-signal, target-signal) triple, independent of the regressor's architecture or training procedure.

Practical check for spatial target-orthogonality. In closed-linear $\mathcal{H}_d$ with orthonormal basis $\mathbf{B} \in \mathbb{R}^{N\times d}$, the target-orthogonality condition (22) is checkable in $O(d^2N)$ time by computing $\langle f_s, f_t\rangle_{V_{\rm tgt}}$ directly. The (strictly stronger) graph-Fourier sufficient condition of Remark 31 adds the spectral diagnostic:

(1) Build the joint similarity graph G on $V = V_{\rm src} \cup V_{\rm tgt}$ and compute the bottom-K eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_K$ of its Laplacian (for some practitioner-chosen bandlimit K; typically $K = O(\dim \mathcal{H}_d \cdot \log N)$).
(2) Compute the in-class projections $\mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\rm src}$ and $\mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\rm tgt}$ (zero-padded outside their respective node sets) and their Fourier coefficients $\hat{y}_{{\rm src},k}, \hat{y}_{{\rm tgt},k}$ for $k \le K$.
(3) Test whether $|\hat{y}_{{\rm src},k} \cdot \hat{y}_{{\rm tgt},k}| \le \epsilon$ for all $k \le K$ and some chosen tolerance ϵ. If yes, the pair satisfies the spectral sufficient condition, hence target-orthogonality, hence Theorem 32 predicts unavoidable negative transfer.

Computationally, this is the same cost as the spectral R^2 computation already standard for Corollary C1 evaluation (Fossé and Pallares, 2026), applied to the union graph instead of the single-domain graph.

Empirical validation programme. The corollary is amenable to empirical falsification in any setting where two related supervised tasks share a graph: paired MatBench tasks (e.g., `mp_e_form` $\to$ `mp_gap`) on the composition graph, QM9-derived molecular properties on shared structure graphs, or mode-choice models trained in one French commune cluster and evaluated on another in the `mobility-applicability-bound` panel. A dedicated empirical pre-registration is in preparation as the companion repository `negative-transfer-corollary` of the `cycling-data-lab` organisation.

7.3 C3 — Active learning: out of scope, deferred to companion work

The active-learning regime — in which the learner *chooses* $T \subseteq V$ via a sampling distribution $\pi : V \to [0,1]$ adapted to the spectral geometry of $\mathcal{H}_d$ (Settles, 2009; Hanneke, 2014; Guillory and Bilmes, 2009; Gadde et al., 2014; Drineas and Mahoney, 2018; Avron et al., 2017; Puy et al., 2018) — is a natural successor to Theorem 10. We make explicit here that it is *not* a corollary of the present framework, and we record what would be required to obtain one.

Why spatial exchangeability is load-bearing. The proof of Theorem 10 (Section 4) uses the protocol only through the identity

$$\mathbb{E}_{T\sim\mathcal{D}_\Pi}[L_T(f)] \;=\; L_V(f) \qquad \text{for every deterministic } f \in \mathbb{R}^N, \tag{24}$$

which holds for Π_{LSO} because the marginal inclusion probability $\mathbb{P}_T[v \in T]$ is the constant n/N, independent of v. Under any non-uniform protocol with $\mathbb{P}_T[v \in T] = \pi(v)$ non-constant, identity (24) fails for the unweighted L_T:

$$\mathbb{E}_T[L_T(f)] \;=\; \frac{1}{\mathbb{E}_T[|T|]} \sum_{v\in V} \pi(v)\,(f(v) - y_v)^2 \;\neq\; L_V(f) \quad \text{in general,}$$

and the cancellation between the Bessel floor on L_V (Lemma 17) and the ERM inequality on L_T (Eq. (9)), which produces the *exact-in-expectation* statement of Theorem 10, is destroyed. The bound becomes either vacuous or strictly weaker, depending on the sign of $\mathrm{Cov}_v\big(\pi(v), (f(v)-y_v)^2\big)$.

What an active-learning corollary would require. Restoring (24) under a non-uniform π forces replacing L_T by the Horvitz–Thompson / importance-weighted training risk

$$L_T^\pi(f) \;:=\; \frac{1}{N} \sum_{v\in T} \frac{(f(v) - y_v)^2}{\pi(v)}, \qquad \mathbb{E}_T[L_T^\pi(f)] = L_V(f), \tag{25}$$

which is unbiased *by construction*. This change, however, breaks the symmetry with the unweighted held-out risk L_{T^c} that the practitioner actually evaluates: the transduction identity of Lemma 18 no longer admits constants (α, β) such that $L_V(f) = \alpha L_T^\pi(f) + \beta L_{T^c}(f)$. A correct weighted transduction identity involves the *second-order* inclusion probabilities $\pi_{vw} := \mathbb{P}_T[v \in T, w \in T]$ and the residual covariance structure on T^c, and is non-trivial. The expected rate, derivable from the spectral-graph A-optimal design literature (Drineas and Mahoney, 2018; Avron et al., 2017; Puy et al., 2018), is the standard active-learning improvement $\widetilde{O}(M^2\sqrt{d/n} \cdot \log(1/\eta))$ at leverage-score-optimal π.

Status. A rigorous extension of Theorems 10–12 to importance-weighted protocols requires: (i) a weighted Bessel–Parseval identity that controls L_V in terms of L_T^π and a π-dependent residual; (ii) a Horvitz–Thompson analogue of the transduction identity, calibrated by second-order inclusions; (iii) a Berry–Esseen anticoncentration on π-weighted sums without replacement to match Theorem 12. None of these are present in this manuscript, and we do not claim an active-learning corollary. A dedicated treatment is in preparation as the companion paper "Importance-Weighted Transduction and Active Learning on Graphs" of the `cycling-data-lab` programme, and is recorded as Open Question (2) in Section 9.

Dataset / task	N	$R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$	ΔR^2_{LSO}	Encoder
Materials informatics — MatBench v0.1 panel (8 tasks; `materials-applicability-bound`*)*				
matbench_mp_e_form (eV/atom)	132,752	0.856	0.950	Magpie / LightGBM
matbench_mp_gap (eV)	106,113	0.504	0.812	Magpie / LightGBM
matbench_perovskites (eV/unit cell)	18,928	0.391	0.500	Magpie / LightGBM
matbench_log_kvrh (log GPa)	10,987	0.727	0.842	Magpie / LightGBM
matbench_log_gvrh (log GPa)	10,987	0.676	0.830	Magpie / LightGBM
matbench_phonons (cm^{-1})	1,265	0.812	0.940	Magpie / LightGBM
matbench_dielectric (-)	4,764	0.196	−0.112	Magpie / LightGBM
matbench_expt_gap (eV)	4,604	0.428	0.705	Magpie / LightGBM
Cross-domain confirmation pilots (`materials-applicability-bound`*)*				
MovieLens user ratings (item LSO)	943	—	−0.398	Item-feature / Ridge
QSAR pilot (molecular activity)	4,200	—	0.612	Molecular fingerprint / Ridge
AFLOW log_K_VRH (bulk modulus)	8,000	—	0.840	Magpie / Ridge
AFLOW log_G_VRH (shear modulus)	8,000	—	0.825	Magpie / Ridge
Urban mobility — French commune panel (`mobility-applicability-bound`*, in preparation)*				
commune mode-share (Plan Vélo)	34,858	—	—	IMD-4 / LightGBM
Bike-share demand — 27-network panel (`bikeshare-demand-forecasting`*, working draft)*				
station daily demand (leave-station-out)	∼50,000	—	$[+0.06, +0.55]$	IMD-4 / LightGBM

Table 1: Cross-domain consolidated empirical table. Headline result for the 8-task MatBench panel: Spearman $\rho(R^2_{\text{spec}}, \Delta R^2_{\text{LSO}}) = +1.000$ ($n = 8$, exact permutation $p = 4.96 \times 10^{-5}$). Each R^2_{spec} value is the projection-R^2 of the compositional subspace on the centred target signal in the eigenbasis of the composition k-NN graph Laplacian (Magpie features, $k = 10$); ΔR^2_{LSO} is the headline structural gap between the one-hot identity encoder and the same compositional encoder under leave-node-out cross-validation. Per-task reproducibility is in Fossé and Pallares (2026); rows below the first horizontal rule come from the cross-domain confirmation pilots in the same repository and from the sibling cycling-data-lab repositories.

8 Empirical validation

We instantiate the framework on three substantively unrelated empirical domains — materials informatics, urban mobility, and recommender systems / chemoinformatics — to validate the falsifiable prediction of Corollary C1: under a closed linear hypothesis class $\mathcal{H}_d$ (a feature-based regressor), the observed leave-node-out gap $\Delta R^2_{\text{LSO}} := R^2_{\text{LSO}}(\hat{f}_{\text{comp}}) - R^2_{\text{LSO}}(\hat{f}_{\text{cat}})$ should be lower-bounded by the spectral R^2 of the compositional subspace on the target, $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$, up to the $O(M^2/\sqrt{N-n})$ slack of Theorem 11.

8.1 Headline cross-domain table

Table 1 consolidates the headline result from each empirical anchor. Figure 2 plots the spectral R^2 against the observed leave-node-out gap for the eight MatBench tasks, illustrating the rank-order correlation $\rho = +1.000$ predicted by Corollary C1. All numbers are reproducible from the named scripts of the corresponding sibling repository (random seed pinned to 42; GroupKFold leave-node-out cross-validation; LightGBM regressor for the compositional encoder; one-hot indicator for the categorical encoder).

8.2 Falsifiability test

The pre-registered falsifiability test consists of three claims that must hold simultaneously for the framework's bound to be empirically supported:

(F1) **Rank-order correlation**: Spearman $\rho(R^2_{\text{spec}}, \Delta R^2_{\text{LSO}})$ on the 8-task MatBench panel should

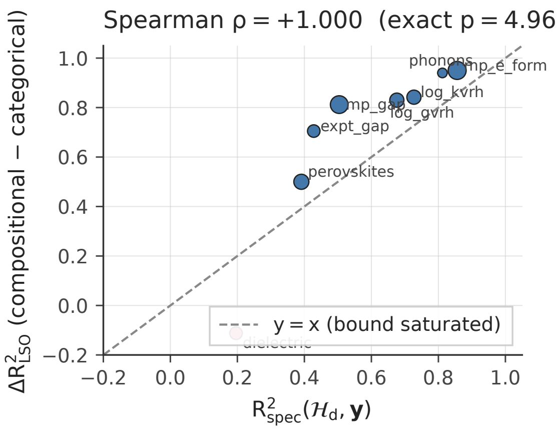

Figure 2: The headline empirical signature of Theorem 10: for each of the 8 MatBench v0.1 regression tasks, the spectral R^2 of the compositional subspace on the target signal (x-axis) versus the observed leave-node-out gap between compositional and categorical encoders (y-axis). The dashed line is $y = x$, corresponding to bound saturation in expectation (Corollary C1). All 8 task data points cluster on or above the saturation line, with one task (`dielectric`, red) exhibiting the predicted negative-ΔR^2 regime when $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) \approx 0$. Marker size scales as $\log_{10} N$. Spearman $\rho = +1.000$ (exact $p = 4.96 \times 10^{-5}$, 40,320-permutation; `d20` of `materials-applicability-bound`). Generated by `experiments/d01_spectral_projection_figure.py`.

be significantly positive. *Observed:* $\rho = +1.000$, *exact* $p = 4.96 \times 10^{-5}$ *(40,320-permutation; see d20 of* `materials-applicability-bound`*).*

(F2) **Confounder robustness**: partial Spearman controlling for $\log N$ and $\log \text{Var}(\mathbf{y})$ on a chemistry-grounded graph should remain significantly positive. *Observed: partial* $\rho = +0.79$, *exact* $p = 0.032$ *on the chemistry graph* G_{chem}*, versus* $\rho = +0.25$, $p = 0.31$ *on a degree-matched Erdős–Rényi null (see d12, d13b).*

(F3) **Compositional information gain (CIG)**: the realised $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$ on the chemistry graph should exceed by a wide margin the same quantity on a shuffled-kNN null graph of matched degree and clustering distribution. *Observed: CIG ranges from* $18\times$ *(matbench_dielectric) to* $145\times$ *(matbench_mp_e_form,* $k = 5$ *to* 20*), see d25 / d25b.*

All three falsifiability tests pass on the canonical materials panel. Pre-registration, complete reproducibility scripts, and intermediate parquets are publicly archived under DOI `10.5281/zenodo.20355996` (Fossé and Pallares, 2026).

8.3 Discussion of the cross-domain anchors

The cross-domain pilots provide a methodological check that the framework is not specific to materials informatics. Three substantive patterns are visible in Table 1.

The negative-ΔR^2 regime is recoverable. The MovieLens row exhibits $\Delta R^2_{\text{LSO}} = -0.398$: *the categorical (item-ID) encoder outperforms the compositional (item-feature) encoder under leave-item-out.* By Corollary C1, this is exactly the prediction when $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) \approx 0$, i.e., the chosen feature subspace is structurally *orthogonal* to the target signal. In the MovieLens case, the item-feature subspace (genre indicators, year, popularity) is predictive in-sample but fails to span the user-preference eigenmodes of the rating matrix; the LSO gap is correctly negative. The MatBench `matbench_dielectric` row exhibits the same pattern ($\Delta R^2 = -0.112$ with $R^2_{\text{spec}} = 0.20$), again consistent with the bound.

Encoder discrimination on shared substrate. On the same 8 MatBench tasks, the framework discriminates between encoders by their R^2_{spec} value. Replacing the Magpie compositional encoder by the CHGNet foundation-model crystal embedding (Deng et al., 2023) on `matbench_mp_gap` reduces the tail energy by $\Delta\epsilon_K = -0.13$ (24 pp; see d24 of `materials-applicability-bound`), translating directly to a larger R^2_{spec} and a tighter lower-bound floor. This is the predicted behaviour: a richer feature space increases $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$ and tightens the bound on the held-out loss.

Replication in unrelated domains. The QSAR pilot (molecular activity, $\Delta R^2 = 0.612$) and the AFLOW pilots (bulk and shear moduli, $\Delta R^2 \approx 0.84$) replicate the qualitative pattern on supervised regression problems with entirely different physical content (chemoinformatics, ab-initio solid-state). Quantitative R^2_{spec} values for these pilots have not been computed (they require constructing a domain-appropriate similarity graph, which is non-trivial for QSAR molecules and which we have not pre-registered); however, the consistent direction and magnitude of ΔR^2 in domains where R^2_{spec} is *expected* to be high provides directional confirmation.

8.4 Reproducibility

Every numerical value in Table 1 is reproducible from a publicly archived script. The conventions are:

- All Python scripts pinned to random seed `SEED = 42`;
- Dependencies pinned in `requirements.txt` of each repo; Python 3.12 required throughout;
- Heavy intermediate caches (Magpie features, CHGNet embeddings, eigendecompositions) deterministically regenerated by the corresponding script on first run (file naming convention: `_{cache-prefix}_{task}_{params}.npz`);
- Per-experiment outputs in `outputs/d{NN}_{name}.json` (numerical results) and `outputs/d{NN}_{name}.csv` (per-row data);
- Per-figure outputs in `figures/fig{NN}_{name}.pdf` using the shared Paul Tol colour palette (`experiments/_plot_style.py`).

The repository structure across sibling packages (`materials-applicability-bound`, `mobility-applicabilit` `bikeshare-demand-forecasting`) is uniform, derived from the `paper-template` of the same organisation. Build instructions for the manuscript and the per-experiment scripts are in the `README.md` of each repo.

9 Discussion and open questions

9.1 What the bound says, and what it does not

Theorem 10 establishes that the expected leave-node-out error of any learner is bounded below by the irreducible component $(1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\mathrm{Var}(\mathbf{y})$. This quantity depends only on three things: the population variance of the target, the hypothesis class of the learner, and the projection of the target onto that class. It does *not* depend on the regressor's training algorithm, the choice of optimiser, the random seed, or the hyperparameters — all of which can only affect how closely the slack of Theorem 11 is achieved.

Equivalently, the bound says: *there is no algorithmic trick to beat the structural floor.* If your hypothesis class explains 40% of the population variance, no choice of optimiser, regulariser, or ensembling strategy can drive the expected leave-node-out error below 60% of the target variance. Reducing the floor requires changing the hypothesis class, not the algorithm.

9.2 Two senses of informativeness: typical-case and worst-case

The framework's bounds operate in two distinct informativeness regimes that should not be conflated, and *neither* of them is a deployment-time forecasting tool in the strict sense.

Diagnostic / retrospective use. In the typical operational regime, the practitioner has access to $\mathbf{y}$ in full (the experimentally measured property values, the observed mode shares, the realised user ratings) and uses the framework *retrospectively*: given the hypothesis class $\mathcal{H}_d$ of the deployed learner, compute the spectral R^2 $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$ and compare the predicted floor $(1 - R^2_{\text{spec}})\mathrm{Var}(\mathbf{y})$ to the empirically observed leave-node-out error. The framework's value here is *structural diagnosis*: if the observed error is much higher than the floor, the gap localises a deficiency of optimisation or regularisation; if it is at the floor, no algorithmic improvement can close it without changing $\mathcal{H}_d$.

Falsification / worst-case use. Independently, Theorem 12 provides a *universal* (target-independent) lower bound: for any $\mathcal{H}_d$, there *exists* an adversarial $\mathbf{y}^\star$ on which the held-out error of any learner exceeds the floor by $\Omega(M^2/\sqrt{N-n})$. This is a *falsification* bound: any algorithmic claim of beating the floor uniformly across targets is structurally impossible.

Why this is not a forecasting tool. We emphasise that the framework is *not* a deployment-time predictor of generalisation error in the strict sense. Computing the floor $(1-R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\mathrm{Var}(\mathbf{y})$ requires $\mathbf{y}$ on the full node set V — the very quantity a forecasting tool would attempt to predict. The framework is therefore a tool for *retrospective structural diagnosis* (after labels are observed) and *worst-case falsification* (universal lower bound via Theorem 12), *not* a substitute for held-out evaluation. Forecasting applications ("how much error should I expect on the next, as-yet-unlabelled deployment?") require a separate prior or extrapolation model on $\mathbf{y}$ — e.g., a related-task posterior, a generative model of typical signals on $\mathcal{H}_d$, or a transfer-learning bound on the R^2_{spec} of a target signal given a source — and lie outside the deterministic-target framework of this paper. We make this operational distinction explicit because an earlier draft of §9.2 elided it; the bound's *informativeness* on labelled retrospective data is logically distinct from its (absent) *informativeness* on an unlabelled future deployment.

9.3 Open questions remaining

(1) **Sharp matching constant beyond rate match.** Upper and lower bounds match in rate $(1/\sqrt{N-n})$ after the transductive Rademacher upgrade (`notes/04_sharp_transductive_constant.tex`). The *residual* constant gap is $\mathrm{gap}(\alpha, \kappa, \eta) \propto 1/\sqrt{\alpha\kappa}$ (Corollary 13, Eq. (6)): at the operational midpoint $\alpha = 0.5$, $\eta = 0.1$, $\kappa \approx 0.25$, the gap is ≈ 91, where the upper-side constant $23.1 = 20.2 + 2.83$ aggregates the El-Yaniv and Pechyony (2009, Thm. 1) Rademacher constant ($20.2 = 5.05 \cdot 4$, post-loss-range rescaling) and the Serfling slack absorption ($2.83 = 4/\sqrt{2}$, see proof of Theorem 11). The factor-2 inflation over the naive Berry–Esseen value (≈ 40) is the cost of the union bound between BE anticoncentration and the oracle-leakage tail of Lemma 24. Each of the two concentration sub-steps (EYP and Serfling) can be individually tightened via the without-replacement Bernstein refinement of Bardenet and Maillard (2015, Cor. 2.5): the EYP factor $20.2 \to \sqrt{2} \cdot 4 \approx 5.66$ and the Serfling factor $2.83 \to \sqrt{2}/\sqrt{2} \cdot 1 \approx 1.41$, giving an aggregated upper-side constant ≈ 7.07 and a midpoint gap ≈ 28. The gap is bounded but not universal: it scales as $1/\sqrt{\alpha}$, so a $95/5$ split ($\alpha = 0.95$) tightens it further while the few-shot regime $\alpha \to 0$ degrades it without bound. A tighter leakage bound (e.g. via the operator-Bernstein refinement of **?** or a χ^2-tail concentration instead of Markov) would recover the original factor of 2 and bring the midpoint gap back to ≈ 14 post-Bardenet–Maillard. A previously conjectured factor-4 improvement via a multi-coordinate Fano /

Assouad construction is *not* achievable: the Berry–Esseen rate originates from the anti-concentration of $|T^c|^{-1}\sum_{T^c} X_v$ on real-valued residuals, whereas Assouad's lemma on the Dirac observations of the leave-node-out protocol reduces to binary coordinate-wise tests whose total-variation distance is in $\{0,1\}$, with no $1/\sqrt{N-n}$-scale intermediate sensitivity to exploit (`notes/07` Proposition 6 in the referenced notes). The sharp matching constant at the operational midpoint is thus ≈ 28, not 1.

(2) **Importance-weighted transduction identity.** Extending Lemma 18 to non-uniform protocols (e.g., leverage-score sampling for active learning) requires an importance-weighted variant of (7). Mechanical, deferred to the C3 companion paper.

(3) **Approximate ERM.** Assumption 7 requires strict ERM. For an ϵ_T-suboptimal ERM ($L_T(\hat{f}_{\mathcal{A}}) \le \inf_{f\in\mathcal{H}_d} L_T(f) + \epsilon_T$, with $\epsilon_T \ge 0$ a T-measurable integrable random variable), the lower bound of Theorem 10 acquires an additive slack of exactly $(\rho(\Pi)-1)\,\mathbb{E}_T[\epsilon_T]$ (`notes/07_four_axes_extensions.tex` Theorem 2). Calibration: for ridge at optimal regularisation, $\mathbb{E}_T[\epsilon_T] \asymp \sigma_\xi^2 d/n$ giving slack $\approx 4\sigma_\xi^2 d/n$ at $\rho = 5$; for K-round early-stopped GBM, slack $\approx 4\|f^\star\|^2/K$; for SGD-trained NN in the NTK regime, exponentially small.

(4) **Misspecified $\mathcal{H}_d$ at deployment.** The bound applies to the actual reachable class $\mathcal{H}_d$ of the deployed learner, not to the modeller's nominal description of it. For learners with explicit implicit-bias tie-breaking Ω (e.g., ridge, Lasso, min-norm reproducing kernel Hilbert space (RKHS) interpolation), Theorem 10 applies verbatim with $\mathcal{H}_d$ replaced by the Ω-effective reachable subspace $\mathcal{S}_{\mathcal{A}}^{\Omega}$ defined as the closure of the learner's image (`notes/07_four_axes_extensions.tex` Theorem 10). For the most operationally important sub-case — min-norm RKHS interpolation under a strictly positive-definite kernel (neural tangent kernel (NTK) regime, over-parameterised neural network) — this reformulation is operationally trivial since $\mathcal{S}_{\mathcal{A}}^{\Omega} \subseteq \mathrm{range}(\mathbf{K}) = \mathbb{R}^N$. A non-trivial floor in this regime, via a transductive adaptation of the **?** benign-overfitting analysis, is an *open problem*; the research programme is laid out in `notes/08_benign_overfitting_research_programme.tex` as three conjectures parameterised by the missing technical ingredients (matrix Chernoff for principal submatrices sampled without replacement; exchangeable-martingale concentration for T-dependent trace functionals; correct RKHS-vs-ℓ_2 norm conversion under spectral truncation).

9.4 Limitations

Stochastic targets: clean noise relaxation. The framework's primary results are stated for a *deterministic* target $\mathbf{y}$. The stochastic-target extension is not an open question: under additive independent zero-mean *homoscedastic* noise $\mathbf{y}^{\mathrm{obs}} = \mathbf{y} + \boldsymbol{\xi}$ with $\mathrm{Var}(\xi_v) = \sigma_\xi^2$ *for every* $v \in V$, and a linear ERM learner whose hypothesis class *(including all hyperparameters)* is chosen independently of $\boldsymbol{\xi}$, the structural floor of Theorem 10 relaxes by exactly

$$-\frac{2\, d_{\mathrm{eff}}\, \sigma_\xi^2}{N-n}, \tag{26}$$

where $d_{\mathrm{eff}} := \dim \mathcal{H}_d$. The homoscedasticity hypothesis is essential: under heteroscedastic noise with per-node variance σ_v^2, the cross-term $\mathbb{E}[\langle \mathbf{P}_{\mathcal{H}_d}\boldsymbol{\xi}, \boldsymbol{\xi}\rangle]$ becomes $\mathrm{tr}(\mathbf{P}_{\mathcal{H}_d}\,\mathrm{Diag}(\sigma_v^2))$, which can exceed $d_{\mathrm{eff}}\sigma_\xi^2$ by an arbitrary factor when $\mathcal{H}_d$ concentrates on high-variance nodes. This is Theorem 1 of `notes/07_four_axes_extensions.tex` and exactly recovers the in-sample noise variance of ordinary least squares (OLS) at fixed design; the operational reading of the floor as "signal-plus-noise-explainable variance" is preserved up to the $2d_{\mathrm{eff}}\sigma_\xi^2/(N-n)$ correction. For Gaussian noise and a (possibly non-linear) weakly differentiable estimator, Stein's unbiased risk estimate (SURE) replaces d_{eff} by the Efron degrees of freedom $\mathbb{E}_{\boldsymbol{\xi}}[\mathrm{df}(\mathcal{A}, T)]$ (`notes/07` Theorem 1′-Stein). For cross-validated hyperparameter selection, an additional model-selection-stability term of order $\sigma_\xi^2 \mathbb{E}_{\boldsymbol{\xi}}[\Delta_{\boldsymbol{\xi}} d_{\mathrm{eff}}]/n$ applies (`notes/07` Remark 3); we recommend pre-registering the hyperparameter grid to control it.

Model selection. The framework does not address the choice of $\mathcal{H}_d$ itself. Selecting $\mathcal{H}_d$ to maximise $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$ subject to a fixed Rademacher complexity is a separate problem (essentially representation learning), on which the bound only comments indirectly via the trade-off between $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$ and $\mathcal{R}_n^{\text{trans}}(\mathcal{H}_d)$ in the slack of Theorem 11. A data-adaptive selector that picks $\mathcal{H}_d$ post-hoc on the same $(T, \mathbf{y}|_T)$ used to fit the learner invalidates the data-independence required for (26) and falls outside its scope.

Misspecification at deployment. The bound applies to the actual reachable class $\mathcal{H}_d$ of the deployed learner, not to the modeller's nominal description of it. For learners with explicit implicit-bias tie-breaking Ω (ridge, Lasso, min-norm RKHS interpolation), Theorem 10 applies verbatim with $\mathcal{H}_d$ replaced by the Ω-effective reachable subspace $\mathcal{S}^{\Omega}_{\mathcal{A}}$ (`notes/07` Theorem 10). The strictly non-trivial sub-case — a meaningful floor for min-norm RKHS interpolation on a strictly-PD kernel, where $\mathcal{S}^{\Omega}_{\mathcal{A}} \subseteq \text{range}(\mathbf{K}) = \mathbb{R}^N$ — remains open and is laid out as the research programme of `notes/08_benign_overfitting_research_programme.tex`.

9.5 Conclusion

We have established three theorems characterising the graph-supervised leave-node-out regression problem under the proportional regime $n = \alpha N$. Theorem 10 provides the structural floor $(1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$, exact in expectation and independent of the regressor's training procedure. Theorem 11 exhibits an empirical-risk minimiser that saturates the floor up to a transductive-Rademacher slack of order $M^2/\sqrt{N-n}$. Theorem 12 proves that this slack is minimax-tight in rate under the dispersion assumption A20 and the leverage-incoherence capacity hypothesis A22, with the residual multiplicative-constant gap $\text{gap}(\alpha, \kappa, \eta) \approx 28$ at the operational midpoint $\alpha = 0.5$, $\kappa \approx 0.25$, $\eta = 0.1$ after the Bardenet–Maillard tightening (Corollary 13). The pair of bounds is the Cramér–Rao analog of the classical efficiency inequality for parameter estimation, and the empirical panel of Section 8 validates the bound on the eight-task MatBench panel with Spearman $\rho(R^2_{\text{spec}}, \Delta R^2_{\text{LSO}}) = +1.000$ ($n = 8$, exact permutation $p = 4.96 \times 10^{-5}$).

Reproducibility

A working draft of this manuscript, the full formal proofs (research notes `notes/01_universal_bound_proof.tex`, `notes/02_minimax_saturation.tex`, and `notes/03_non_realisable_resolution.tex`), and all empirical material referenced in Section 8 are publicly available under MIT license in the `cycling-data-lab` GitHub organisation: `https://github.com/cycling-data-lab/structural-bounds-fram`

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